

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 96 on 17/06/2026. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNjA2MTcxODQ5MjVfMjIyMg%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

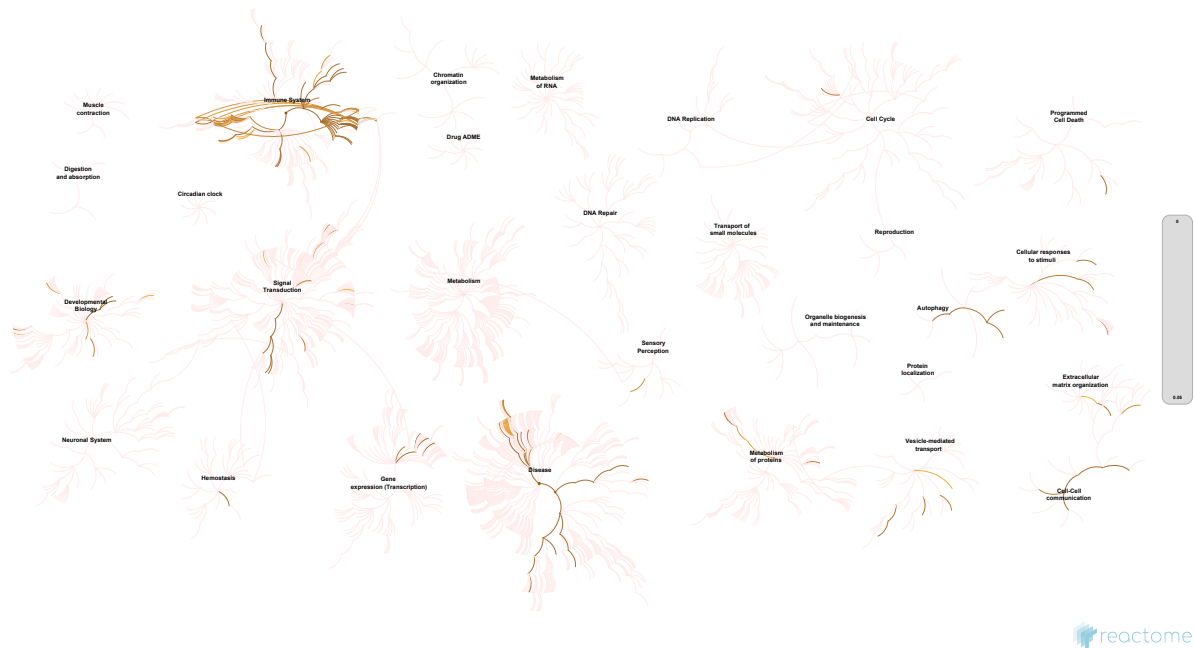
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 100 out of 110 identifiers in the sample were found in Reactome, where 623 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNjA2MTcxODQ5MjVfMjlyMg%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

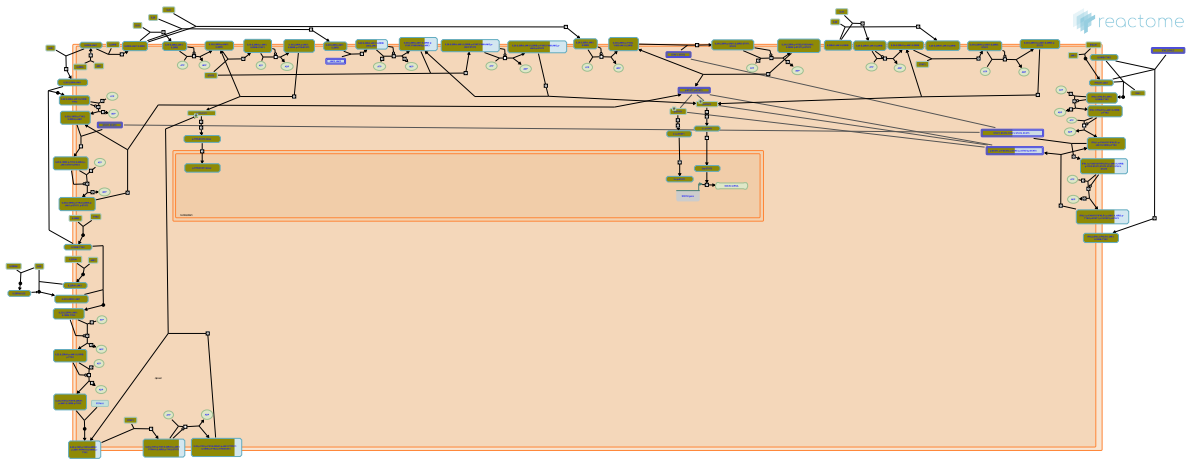
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Interleukin-20 family signaling	19 / 29	0.002	1.11e-16	9.44e-15	56 / 56	0.003
Interleukin-6 signaling	11 / 17	0.001	1.11e-16	9.44e-15	20 / 20	0.001
Interleukin-6 family signaling	13 / 30	0.002	1.11e-16	9.44e-15	33 / 34	0.002
Interleukin-12 family signaling	20 / 96	0.006	1.11e-16	9.44e-15	107 / 114	0.007
Interleukin-4 and Interleukin-13 signaling	31 / 211	0.013	1.11e-16	9.44e-15	42 / 47	0.003
Signaling by Interleukins	75 / 646	0.04	1.11e-16	9.44e-15	361 / 505	0.032
Cytokine Signaling in Immune system	90 / 1,106	0.068	1.11e-16	9.44e-15	454 / 806	0.05
Immune System	105 / 2,810	0.174	1.11e-16	9.44e-15	553 / 1,842	0.115
Interleukin-10 signaling	16 / 86	0.005	5.55e-16	4.16e-14	15 / 15	9.37e-04
Interleukin-35 Signalling	10 / 16	9.88e-04	1.33e-15	9.06e-14	26 / 26	0.002
Regulation of IFNA/IFNB signaling	11 / 28	0.002	7.66e-15	4.67e-13	5 / 5	3.12e-04
Activation of STAT3 by cadherin engagement	14 / 87	0.005	2.61e-13	1.46e-11	21 / 33	0.002
SARS-CoV-2 activates/modulates innate and adaptive immune responses	17 / 160	0.01	4.87e-13	2.53e-11	6 / 48	0.003
Differentiation of T cells	14 / 100	0.006	1.65e-12	7.92e-11	16 / 28	0.002
Interleukin-2 family signaling	11 / 47	0.003	1.95e-12	8.79e-11	50 / 59	0.004
Interleukin-12 signaling	13 / 84	0.005	3.09e-12	1.30e-10	56 / 56	0.003
Evasion by RSV of host interferon responses	10 / 37	0.002	5.03e-12	2.01e-10	3 / 12	7.50e-04
Interleukin-23 signaling	7 / 11	6.79e-04	2.38e-11	8.81e-10	12 / 12	7.50e-04
Interferon alpha/beta signaling	14 / 129	0.008	4.60e-11	1.61e-09	24 / 25	0.002
SARS-CoV-2-host interactions	17 / 248	0.015	4.09e-10	1.39e-08	6 / 68	0.004
DDX58/IFIH1-mediated induction of interferon-alpha/beta	12 / 105	0.006	6.57e-10	2.02e-08	10 / 53	0.003
TRAF6 mediated IRF7 activation	9 / 44	0.003	6.73e-10	2.02e-08	4 / 13	8.12e-04
Differentiation of naive CD4+ T cells to T helper 1 cells (Th1 cells)	7 / 20	0.001	1.46e-09	4.22e-08	8 / 16	1.00e-03
Interleukin-18 signaling	6 / 11	6.79e-04	1.64e-09	4.59e-08	4 / 4	2.50e-04
MAPK1 (ERK2) activation	6 / 12	7.41e-04	2.74e-09	7.39e-08	1 / 3	1.87e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Interleukin-20 family signaling (R-HSA-8854691)



The interleukin 20 (IL20) subfamily comprises IL19, IL20, IL22, IL24 and IL26. They are members of the larger IL10 family, but have been grouped together based on their usage of common receptor subunits and similarities in their target cell profiles and biological functions. Members of the IL20 subfamily facilitate the communication between leukocytes and epithelial cells, thereby enhancing innate defence mechanisms and tissue repair processes at epithelial surfaces. Much of the understanding of this group of cytokines is based on IL22, which is the most studied member (Rutz et al. 2014, Akdis M et al. 2016, Longsdon et al. 2012).

References

Rutz S, Wang X & Ouyang W (2014). The IL-20 subfamily of cytokines--from host defence to tissue homeostasis. *Nat. Rev. Immunol.*, 14, 783-95. [🔗](#)

Logsdon NJ, Deshpande A, Harris BD, Rajashankar KR & Walter MR (2012). Structural basis for receptor sharing and activation by interleukin-20 receptor-2 (IL-20R2) binding cytokines. *Proc. Natl. Acad. Sci. U.S.A.*, 109, 12704-9. [🔗](#)

Akdis M, Aab A, Altunbulakli C, Azkur K, Costa RA, Cramer R, ... Akdis CA (2016). Interleukins (from IL-1 to IL-38), interferons, transforming growth factor β , and TNF- β : Receptors, functions, and roles in diseases. *J. Allergy Clin. Immunol.*, 138, 984-1010. [🔗](#)

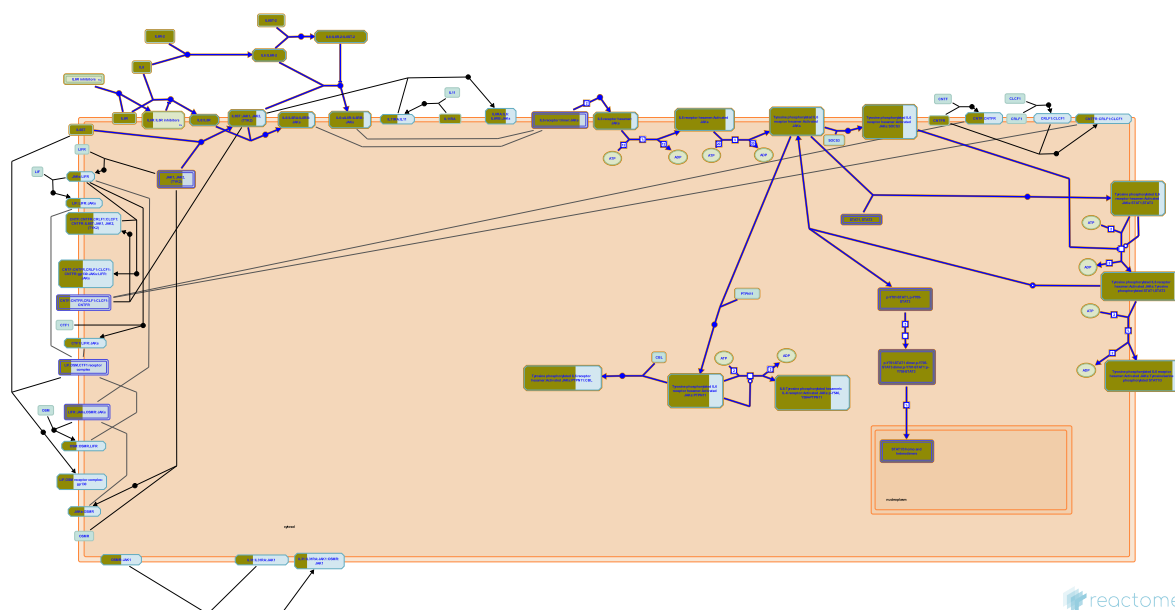
Edit history

Date	Action	Author
2014-06-04	Authored	Jupe S
2016-01-28	Edited	Jupe S
2016-01-28	Reviewed	Meldal BH
2016-01-28	Created	Jupe S
2017-11-15	Reviewed	Datta SK

21 submitted entities found in this pathway, mapping to 25 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IFNL1	Q8IU54	IFNL2	Q8IZI9, Q8IZJ0	IFNL3	Q8IZI9, Q8IZJ0
IFNLR1	Q8IU57, Q969J5	IL10	Q9NPH9	IL10RA	Q8IU57
IL10RB	Q08334	IL19	Q9UHD0	IL20	Q9NYY1
IL20RA	Q9UHF4	IL20RB	Q6UXL0	IL22	Q9GZX6
IL22RA1	Q8N6P7	IL22RA2	Q8N6P7, Q969J5	IL24	Q13007
IL26	Q9NPH9	JAK1	P23458	STAT1	P42224
STAT3	P40763	STAT4	Q14765	TYK2	P29597

2. Interleukin-6 signaling (R-HSA-1059683)



Interleukin-6 (IL-6) is a pleiotropic cytokine with roles in processes including immune regulation, hematopoiesis, inflammation, oncogenesis, metabolic control and sleep. It is the founding member of a family of IL-6-related cytokines such as IL-11, IL-27 leukemia inhibitory factor (LIF), ciliary neurotrophic factor (CNTF) and oncostatin M.

The IL-6 receptor (IL6R) consists of an alpha subunit that specifically binds IL-6 and a beta subunit, IL6RB or gp130, which is the signaling component of all the receptors for cytokines related to IL-6. IL6R alpha exists in transmembrane and soluble forms. The transmembrane form is mainly expressed by hepatocytes, neutrophils, monocytes/macrophages, and some lymphocytes. Soluble forms of IL6R (sIL6R) are also expressed by these cells. Two major mechanisms for the production of sIL6R have been proposed. Alternative splicing generates a transcript lacking the transmembrane domain by using splicing donor and acceptor sites that flank the transmembrane domain coding region. This also introduces a frameshift leading to the incorporation of 10 additional amino acids at the C terminus of sIL6R. A second mechanism for the generation of sIL6R is the proteolytic cleavage or 'shedding' of membrane-bound IL-6R. Two proteases ADAM10 and ADAM17 are thought to contribute to this (Briso et al. 2008). sIL6R can bind IL6 and stimulate cells that express gp130 but not IL6R alpha, a process that is termed trans-signaling. This explains why many cells, including hematopoietic progenitor cells, neuronal cells, endothelial cells, smooth muscle cells, and embryonic stem cells, do not respond to IL6 alone, but show a remarkable response to IL6/sIL6R. It is clear that the trans-signaling pathway is responsible for the pro-inflammatory activities of IL-6 whereas the membrane bound receptor governs regenerative and anti-inflammatory IL-6 activities

IL6R signal transduction is mediated by two pathways: the JAK-STAT (Janus family tyrosine kinase-signal transducer and activator of transcription) pathway and the Ras-MAPK (mitogen-activated protein kinase) pathway. Negative regulators of IL-6 signaling include SOCS (suppressor of cytokine signals) and SHP2. Within the last few years different antibodies have been developed to inhibit IL-6 activity, and the first such antibodies have been introduced into the clinic for the treatment of inflammatory diseases (Kopf et al. 2010).

References

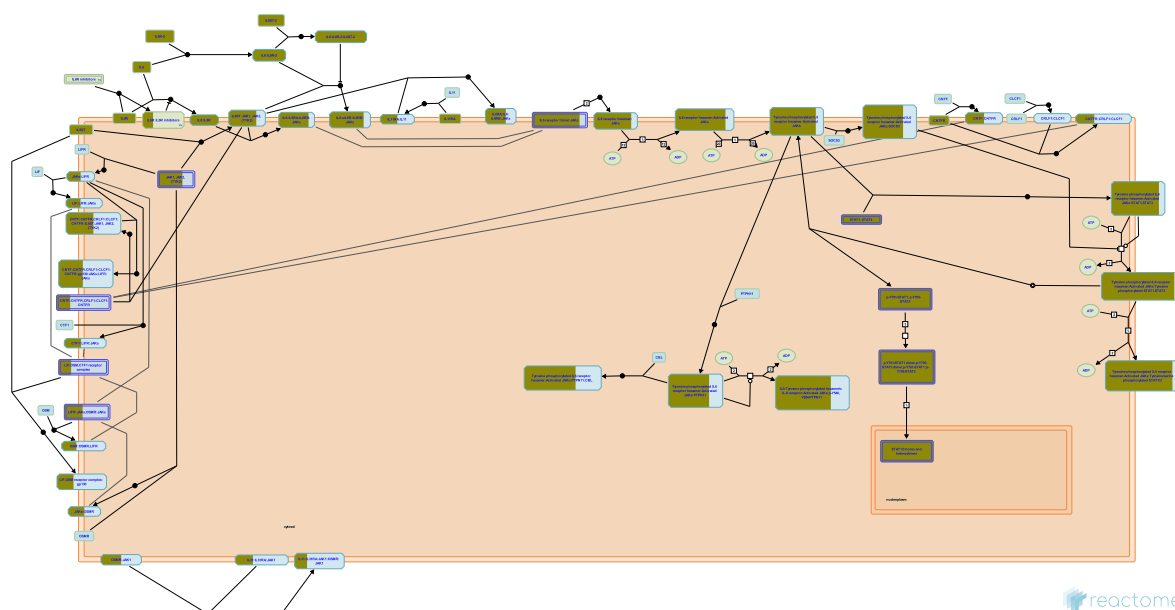
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Date	Action	Author
2010-11-25	Created	Jupe S
2010-12-10	Edited	Jupe S
2010-12-13	Authored	Ray KP
2011-02-11	Reviewed	Rose-John S

8 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL11RA	P08887	IL6	P05231	IL6R	P08887, P08887-2
IL6ST	P40189, P40189-1, P40189-2	JAK1	P23458	STAT1	P42224, P42224-1
STAT3	P40763	TYK2	P29597		

3. Interleukin-6 family signaling (R-HSA-6783589)



Cellular compartments: plasma membrane, extracellular region.

The interleukin-6 (IL6) family of cytokines includes IL6, IL11, IL27, leukemia inhibitory factor (LIF), oncostatin M (OSM), ciliary neurotrophic factor (CNTF), cardiotrophin 1 and 2 (CT-1) and cardiotrophin-like cytokine (CLC) (Heinrich et al. 2003, Pflanz et al. 2002). The latest addition to this family is IL31, discovered in 2004 (Dillon et al. 2004). The family is defined largely by the shared use of the common signal transducing receptor Interleukin-6 receptor subunit beta (IL6ST, gp130). The IL31 receptor uniquely does not include this subunit, instead it uses the related IL31RA. The members of the IL6 family share very low sequence homology but are structurally highly related, forming anti-parallel four-helix bundles with a characteristic “up-up-down-down” topology (Rozwarski et al. 1994, Cornelissen et al. 2012).

Although each member of the IL6 family signals through a distinct receptor complex, their underlying signaling mechanisms are similar. Assembly of the receptor complex is followed by activation of receptor-associated Janus kinases (JAKs), believed to be constitutively associated with the receptor subunits. Activation of JAKs initiates downstream cytoplasmic signaling cascades that involve recruitment and phosphorylation of transcription factors of the Signal transducer and activator of transcription (STAT) family, which dimerize and translocate to the nucleus where they bind enhancer elements of target genes leading to transcriptional activation (Nakashima & Taga 1998).

Negative regulators of IL6 signaling include Suppressor of cytokine signals (SOCS) family members and PTPN11 (SHP-2).

IL6 is a pleiotropic cytokine with roles in processes including immune regulation, hematopoiesis, inflammation, oncogenesis, metabolic control and sleep.

IL6 and IL11 bind their corresponding specific receptors IL6R and IL11R respectively, resulting in dimeric complexes that subsequently associate with IL6ST, leading to IL6ST homodimer formation (in a hexameric or higher order complex) and signal initiation. IL6R alpha exists in transmembrane and soluble forms. The transmembrane form is mainly expressed by hepatocytes, neutrophils, monocytes/macrophages, and some lymphocytes. Soluble forms of IL6R (sIL6R) are also expressed by these cells. Two major mechanisms for the production of sIL6R have been proposed. Alternative splicing generates a transcript lacking the transmembrane domain by using splicing donor and acceptor sites that flank the transmembrane domain coding region. This also introduces a frameshift leading to the incorporation of 10 additional amino acids at the C terminus of sIL6R. A second mechanism for the generation of sIL6R is the proteolytic cleavage or 'shedding' of membrane-bound IL-6R. Two proteases ADAM10 and ADAM17 are thought to contribute to this (Briso et al. 2008). sIL6R can bind IL6 and stimulate cells that express gp130 but not IL6R alpha, a process that is termed trans-signaling. This explains why many cells, including hematopoietic progenitor cells, neuronal cells, endothelial cells, smooth muscle cells, and embryonic stem cells, do not respond to IL6 alone, but show a remarkable response to IL6/sIL6R. It is clear that the trans-signaling pathway is responsible for the pro-inflammatory activities of IL6 whereas the membrane bound receptor governs regenerative and anti-inflammatory IL6 activities

LIF, CNTF, OSM, CTF1, CRLF1 and CLCF1 signal via IL6ST:LIFR heterodimeric receptor complexes (Taga & Kishimoto 1997, Mousa & Bakhiet 2013). OSM signals via a receptor complex consisting of IL6ST and OSMR. These cytokines play important roles in the regulation of complex cellular processes such as gene activation, proliferation and differentiation (Heinrich et al. 1998).

Antibodies have been developed to inhibit IL6 activity for the treatment of inflammatory diseases (Kopf et al. 2010).

References

- Taga T & Kishimoto T (1997). Gp130 and the interleukin-6 family of cytokines. *Annu. Rev. Immunol.*, 15, 797-819. [↗](#)
- Taga T (1996). Gp130, a shared signal transducing receptor component for hematopoietic and neuropoietic cytokines. *J. Neurochem.*, 67, 1-10. [↗](#)
- Kishimoto T, Akira S, Narazaki M & Taga T (1995). Interleukin-6 family of cytokines and gp130. *Blood*, 86, 1243-54. [↗](#)
- Heinrich PC, Behrmann I, Müller-Newen G, Schaper F & Graeve L (1998). Interleukin-6-type cytokine signalling through the gp130/Jak/STAT pathway. *Biochem. J.*, 334, 297-314. [↗](#)
- Nakashima K & Taga T (1998). gp130 and the IL-6 family of cytokines: signaling mechanisms and thrombopoietic activities. *Semin. Hematol.*, 35, 210-21. [↗](#)

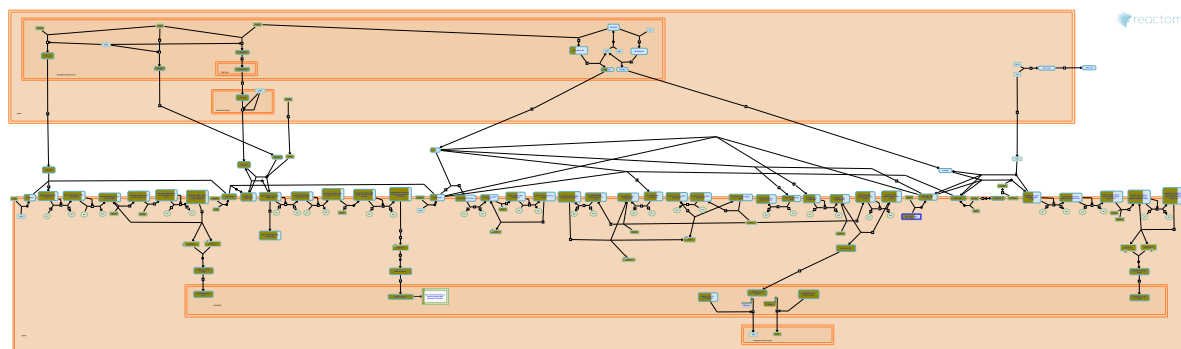
Edit history

Date	Action	Author
2015-06-16	Edited	Garapati P V
2015-06-16	Authored	Garapati P V
2015-06-16	Created	Garapati P V
2015-11-09	Reviewed	Narazaki M, Tanaka M
2026-03-16	Modified	Weiser JD

9 submitted entities found in this pathway, mapping to 14 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CNTFR	P26992	IL11RA	P08887, Q14626	IL6	P05231
IL6R	P08887, P08887-2	IL6ST	P40189, P40189-1, P40189-2	JAK1	P23458
STAT1	P42224, P42224-1	STAT3	P40763	TYK2	P29597

4. Interleukin-12 family signaling ([R-HSA-447115](#))



Interleukin-12 (IL-12) is a heterodimer of interleukin-12 subunit alpha (IL12A, IL-12p35) and interleukin-12 subunit beta (IL12B, IL-12p40). It is a potent immunoregulatory cytokine involved in the generation of cell mediated immunity to intracellular pathogens. It is produced by antigen presenting cells, including dendritic cells, macrophages/monocytes, neutrophils and some B cells (D'Andrea et al. 1992, Kobayashi et al.1989, Heufler et al.1996). It enhances the cytotoxic activity of natural killer (NK) cells and cytotoxic T cells, stimulating proliferation of activated NK and T cells and induces production of interferon gamma (IFN gamma) by these cells (Stern et al. 1990). IL-12 also plays an important role in immunomodulation by promoting cell mediated immunity through induction of a class 1 T helper cell (Th1) immune response. IL-12 may contribute to immunopathological conditions such as rheumatoid arthritis (McIntyre et al. 1996).

The receptor for IL-12 is a heterodimer of IL-12Rbeta1 (IL12RB1) and IL-12Rbeta2 (IL12RB2), both highly homologous to Interleukin-6 receptor subunit beta (IL6ST, gp130). Each has an extracellular ligand binding domain, a transmembrane domain and a cytosolic domain containing box 1 and box 2 sequences that mediate binding of Janus family tyrosine kinases (JAKs). IL-12 binding is believed to bring about the heterodimerization and generation of a high affinity receptor complex capable of signal transduction. In this model, receptor dimerization leads to juxtaposition of the cytosolic domains and subsequent tyrosine phosphorylation and activation of JAK2 and TYK2. These activated kinases, in turn, tyrosine phosphorylate and activate several members of the signal transducer and activator of transcription (STAT) family, mainly STAT4, while also STAT1, STAT3 and STAT5 have been reported to be activated (Bacon et al. 1995, Jacobson et al. 1995, Yu et al. 1996, Gollob et al.1995). The STATs translocate to the nucleus to activate transcription of several genes, including IFN gamma. The production of IFN gamma has a pleiotropic effect in the cell, stimulating production of molecules important to cell mediated immunity. In particular, IFN gamma stimulates production of more IL-12 and sets up a positive regulation loop between IL-12 signaling and IFN gamma (Chan et al. 1991). The importance of IL-12 for this loop is demonstrated by IL-12 and STAT4 knockout mice that are severely compromised in IFN-gamma production (Kaplan et al. 1996; Magram et al. 1996), as well as by patients with IL12B mutations that are severely compromised in IFN-gamma production (Altare et al.1998).

References

- D'Andrea A, Rengaraju M, Valiante NM, Chehimi J, Kubin M, Aste M, ... Nickbarg E (1992). Production of natural killer cell stimulatory factor (interleukin 12) by peripheral blood mononuclear cells. *J Exp Med*, 176, 1387-98. [🔗](#)
- Sun L, He C, Nair L, Yeung J & Egwuagu CE (2015). Interleukin 12 (IL-12) family cytokines: Role in immune pathogenesis and treatment of CNS autoimmune disease. *Cytokine*, 75, 249-55. [🔗](#)

Behzadi P, Behzadi E & Ranjbar R (2016). IL-12 Family Cytokines: General Characteristics, Pathogenic Microorganisms, Receptors, and Signalling Pathways. *Acta Microbiol Immunol Hung*, 63, 1-25. [🔗](#)

Edit history

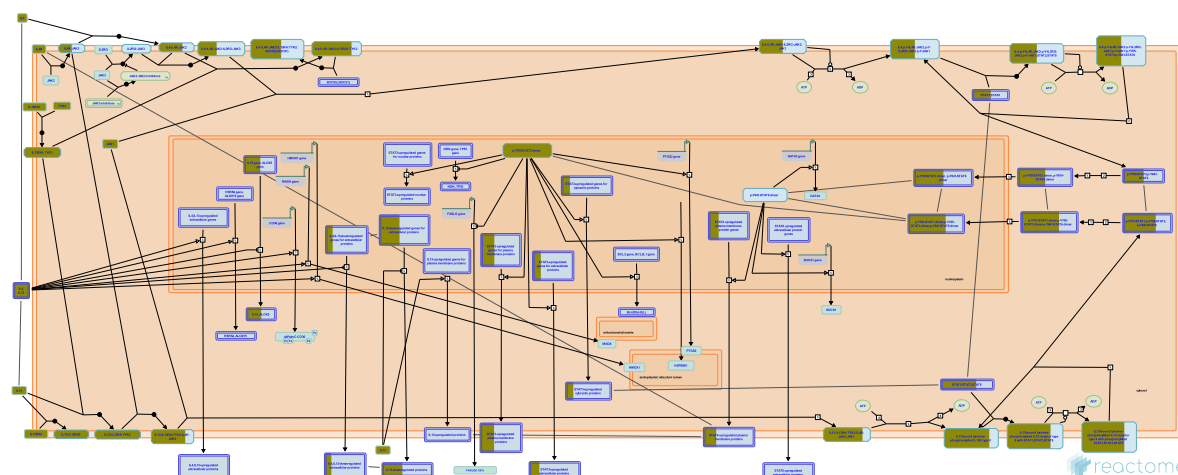
Date	Action	Author
2009-11-20	Created	Jupe S
2014-06-04	Authored	Jupe S
2016-01-28	Edited	Jupe S
2016-01-28	Reviewed	Meldal BH
2026-03-16	Modified	Weiser JD

17 submitted entities found in this pathway, mapping to 21 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CFL1	P23528	IFNG	P01579	IL10	P22301
IL12A	P29459	IL12B	P29460	IL12RB1	P42701
IL12RB2	Q99665	IL23A	Q9NPF7	IL23R	Q5VWK5
IL27RA	Q6UWB1	IL6ST	P40189	JAK1	P23458
PRLR	Q5VWK5	STAT1	P42224	STAT3	P40763
STAT4	Q14765	TYK2	P29597		

Input	Ensembl Id	Input	Ensembl Id
CFL1	ENSG00000172757	IFNG	ENSG00000111537
IL10	ENSG00000136634	IL12A	ENSG00000168811

5. Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)



Interleukin-4 (IL4) is a principal regulatory cytokine during the immune response, crucially important in allergy and asthma (Nelms et al. 1999). When resting T cells are antigen-activated and expand in response to Interleukin-2 (IL2), they can differentiate as Type 1 (Th1) or Type 2 (Th2) T helper cells. The outcome is influenced by IL4. Th2 cells secrete IL4, which both stimulates Th2 in an autocrine fashion and acts as a potent B cell growth factor to promote humoral immunity (Nelms et al. 1999).

Interleukin-13 (IL13) is an immunoregulatory cytokine secreted predominantly by activated Th2 cells. It is a key mediator in the pathogenesis of allergic inflammation. IL13 shares many functional properties with IL4, stemming from the fact that they share a common receptor subunit. IL13 receptors are expressed on human B cells, basophils, eosinophils, mast cells, endothelial cells, fibroblasts, monocytes, macrophages, respiratory epithelial cells, and smooth muscle cells, but unlike IL4, not T cells. Thus IL13 does not appear to be important in the initial differentiation of CD4 T cells into Th2 cells, rather it is important in the effector phase of allergic inflammation (Hershey et al. 2003).

IL4 and IL13 induce “alternative activation” of macrophages, inducing an anti-inflammatory phenotype by signaling through IL4R alpha in a STAT6 dependent manner. This signaling plays an important role in the Th2 response, mediating anti-parasitic effects and aiding wound healing (Gordon & Martinez 2010, Loke et al. 2002)

There are two types of IL4 receptor complex (Andrews et al. 2006). Type I IL4R (IL4R1) is predominantly expressed on the surface of hematopoietic cells and consists of IL4R and IL2RG, the common gamma chain. Type II IL4R (IL4R2) is predominantly expressed on the surface of nonhematopoietic cells, it consists of IL4R and IL13RA1 and is also the type II receptor for IL13. (Obiri et al. 1995, Aman et al. 1996, Hilton et al. 1996, Miloux et al. 1997, Zhang et al. 1997). The second receptor for IL13 consists of IL4R and Interleukin-13 receptor alpha 2 (IL13RA2), sometimes called Interleukin-13 binding protein (IL13BP). It has a high affinity receptor for IL13 (Kd = 250 pmol/L) but is not sufficient to render cells responsive to IL13, even in the presence of IL4R (Donaldson et al. 1998). It is reported to exist in soluble form (Zhang et al. 1997) and when overexpressed reduces JAK-STAT signaling (Kawakami et al. 2001). It's function may be to prevent IL13 signalling via the functional IL4R:IL13RA1 receptor. IL13RA2 is overexpressed and enhances cell invasion in some human cancers (Joshi & Puri 2012).

The first step in the formation of IL4R1 (IL4:IL4R:IL2RB) is the binding of IL4 with IL4R (Hoffman et al. 1995, Shen et al. 1996, Hage et al. 1999). This is also the first step in formation of IL4R2 (IL4:IL4R:IL13RA1). After the initial binding of IL4 and IL4R, IL2RB binds (LaPorte et al. 2008), to form IL4R1. Alternatively, IL13RA1 binds, forming IL4R2. In contrast, the type II IL13 complex (IL13R2) forms with IL13 first binding to IL13RA1 followed by recruitment of IL4R (Wang et al. 2009).

Crystal structures of the IL4:IL4R:IL2RG, IL4:IL4R:IL13RA1 and IL13:IL4R:IL13RA1 complexes have been determined (LaPorte et al. 2008). Consistent with these structures, in monocytes IL4R is tyrosine phosphorylated in response to both IL4 and IL13 (Roy et al. 2002, Gordon & Martinez 2010) while IL13RA1 phosphorylation is induced only by IL13 (Roy et al. 2002, LaPorte et al. 2008) and IL2RG phosphorylation is induced only by IL4 (Roy et al. 2002).

Both IL4 receptor complexes signal through Jak/STAT cascades. IL4R is constitutively-associated with JAK2 (Roy et al. 2002) and associates with JAK1 following binding of IL4 (Yin et al. 1994) or IL13 (Roy et al. 2002). IL2RG constitutively associates with JAK3 (Boussiotis et al. 1994, Russell et al. 1994). IL13RA1 constitutively associates with TYK2 (Umeshita-Suyama et al. 2000, Roy et al. 2002, LaPorte et al. 2008, Bhattacharjee et al. 2013).

IL4 binding to IL4R1 leads to phosphorylation of JAK1 (but not JAK2) and STAT6 activation (Takeda et al. 1994, Ratthe et al. 2007, Bhattacharjee et al. 2013).

IL13 binding increases activating tyrosine-99 phosphorylation of IL13RA1 but not that of IL2RG. IL4 binding to IL2RG leads to its tyrosine phosphorylation (Roy et al. 2002). IL13 binding to IL4R2 leads to TYK2 and JAK2 (but not JAK1) phosphorylation (Roy & Cathcart 1998, Roy et al. 2002).

Phosphorylated TYK2 binds and phosphorylates STAT6 and possibly STAT1 (Bhattacharjee et al. 2013).

A second mechanism of signal transduction activated by IL4 and IL13 leads to the insulin receptor substrate (IRS) family (Kelly-Welch et al. 2003). IL4R1 associates with insulin receptor substrate 2 and activates the PI3K/Akt and Ras/MEK/Erk pathways involved in cell proliferation, survival and translational control. IL4R2 does not associate with insulin receptor substrate 2 and consequently the PI3K/Akt and Ras/MEK/Erk pathways are not activated (Busch-Dienstfertig & González-Rodríguez 2013).

References

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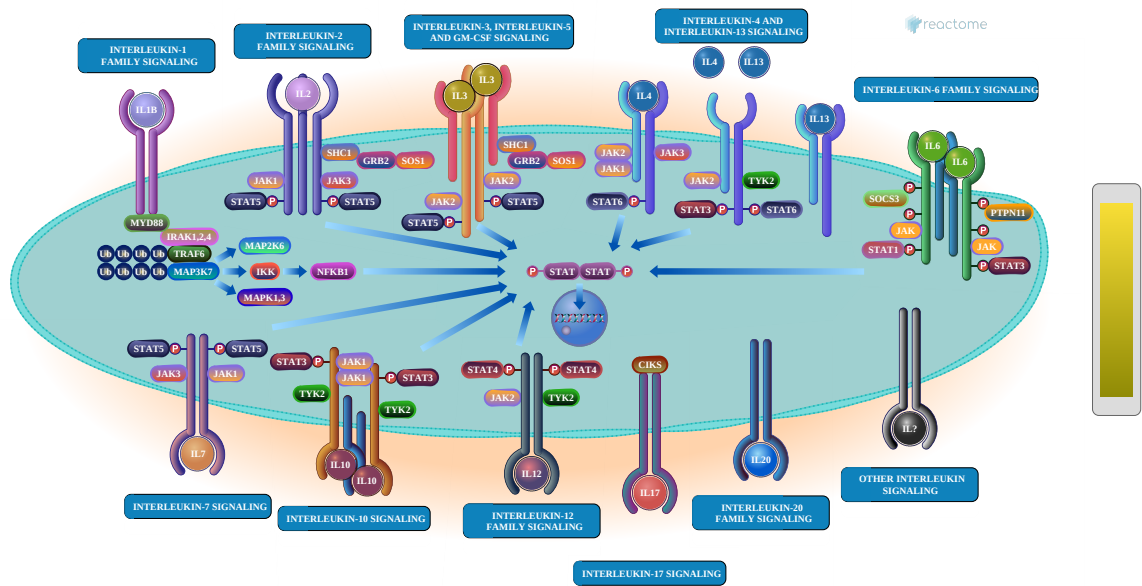
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2015-07-01	Authored	Jupe S
2015-07-01	Created	Jupe S
2016-09-02	Edited	Jupe S
2016-09-02	Reviewed	Leibovich SJ

21 submitted entities found in this pathway, mapping to 33 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL10	P22301	IL11RA	P08887	IL12A	P29459
IL12B	P29460	IL13	P35225	IL13RA1	P78552
IL13RA2	Q14627	IL17A	Q16552	IL18	Q14116
IL23A	Q9NPF7	IL23R	Q5VWK5	IL4	P05112
IL4R	P24394	IL6	P05231	IL6R	P08887
JAK1	P23458	PRLR	Q5VWK5	STAT1	P42224
STAT3	P40763	TYK2	P29597	VIM	P08670

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IL10	ENSG00000136634	IL12A	ENSG00000168811	IL12B	ENSG00000113302
IL17A	ENSG00000112115	IL18	ENSG00000150782	IL23A	ENSG00000110944
IL23R	ENSG00000162594	IL4R	ENSG00000077238	IL6	ENSG00000136244
IL6R	ENSG00000160712	STAT1	ENSG00000115415	VIM	ENSG00000026025

6. Signaling by Interleukins (R-HSA-449147)



Cellular compartments: plasma membrane.

Interleukins are low molecular weight proteins that bind to cell surface receptors and act in an autocrine and/or paracrine fashion. They were first identified as factors produced by leukocytes but are now known to be produced by many other cells throughout the body. They have pleiotropic effects on cells which bind them, impacting processes such as tissue growth and repair, hematopoietic homeostasis, and multiple levels of the host defense against pathogens where they are an essential part of the immune system.

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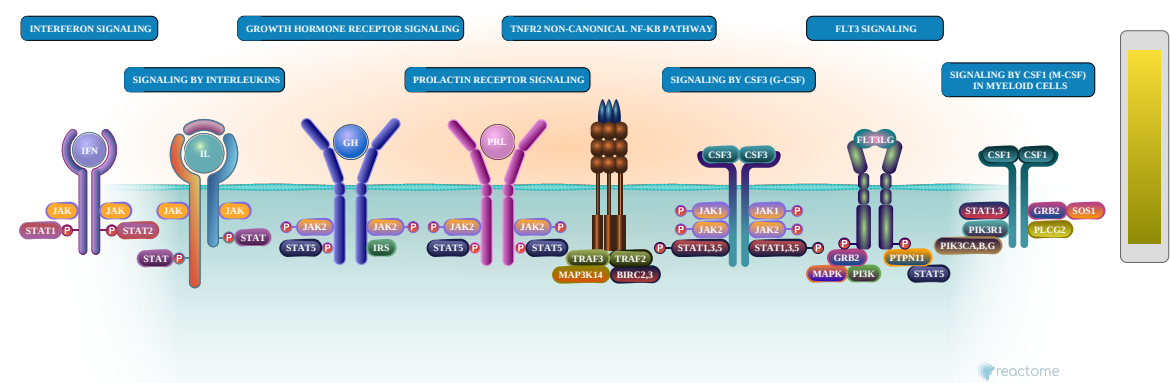
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2009-11-27	Created	Jupe S
2010-05-17	Reviewed	Pinteaux E
2010-05-17	Authored	Ray KP
2010-05-26	Edited	Jupe S
2026-03-16	Modified	Weiser JD

56 submitted entities found in this pathway, mapping to 84 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BTRC	Q9Y297	CFL1	P23528	CNTFR	P26992
CSF2	P04141	IFNG	P01579	IFNL1	Q8IU54
IFNL2	Q8IZI9, Q8IZJ0	IFNL3	Q8IZI9, Q8IZJ0	IFNLR1	Q13651, Q8IU57, Q969J5
IL10	P22301, Q9NPH9	IL10RA	Q13651, Q8IU57	IL10RB	Q08334
IL11RA	P08887, Q14626	IL12A	P29459	IL12B	P29460
IL12RB1	P42701	IL12RB2	Q99665	IL13	P35225
IL13RA1	P78552	IL13RA2	Q14627	IL17A	Q16552
IL18	Q14116, Q14116-2	IL19	Q9UHD0	IL20	Q9NYY1
IL20RA	Q9UHF4	IL20RB	Q6UXL0	IL21R	Q9HBE5
IL22	Q9GZX6	IL22RA1	Q8N6P7	IL22RA2	Q8N6P7, Q969J5
IL23A	Q9NPF7	IL23R	Q5VWK5	IL24	Q13007
IL26	Q9NPH9	IL27RA	Q6UWB1	IL2RA	P01589
IL2RB	P14784	IL37	Q9NZH6	IL4	P05112
IL4R	P24394	IL5RA	Q01344	IL6	P05231
IL6R	P08887, P08887-2	IL6ST	P40189, P40189-1, P40189-2	IL9	P15248
IL9R	Q01113	JAK1	P23458	NOD2	Q9HC29
PRLR	Q5VWK5	RELA	Q04206	RIPK2	O43353
STAT1	P42224, P42224-1	STAT3	P40763	STAT4	Q14765
TYK2	P29597	VIM	P08670		

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CFL1	ENSG00000172757	CSF2	ENSG00000164400	IFNG	ENSG00000111537
IL10	ENSG00000136634	IL12A	ENSG00000168811	IL12B	ENSG00000113302
IL17A	ENSG00000112115	IL18	ENSG00000150782	IL23A	ENSG00000110944
IL23R	ENSG00000162594	IL4R	ENSG00000077238	IL6	ENSG00000136244
IL6R	ENSG00000160712	STAT1	ENSG00000115415	VIM	ENSG00000026025

7. Cytokine Signaling in Immune system (R-HSA-1280215)



Cytokines are small proteins that regulate and mediate immunity, inflammation, and hematopoiesis. They are secreted in response to immune stimuli, and usually act briefly, locally, at very low concentrations. Cytokines bind to specific membrane receptors, which then signal the cell via second messengers, to regulate cellular activity.

References

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COPE. Retrieved from <http://www.copewithcytokines.org/cope.cgi>

Santamaria P (2003). Cytokines and chemokines in autoimmune disease: an overview. *Adv Exp Med Biol*, 520, 1-7.

Edit history

Date	Action	Author
2011-05-12	Created	Garapati P V
2011-05-22	Edited	Ray KP, Jupe S, Garapati P V
2011-05-22	Authored	Ray KP, Jupe S, Garapati P V
2011-05-29	Reviewed	Abdul-Sater AA, Schindler C, Pinteaux E
2026-03-16	Modified	Weiser JD

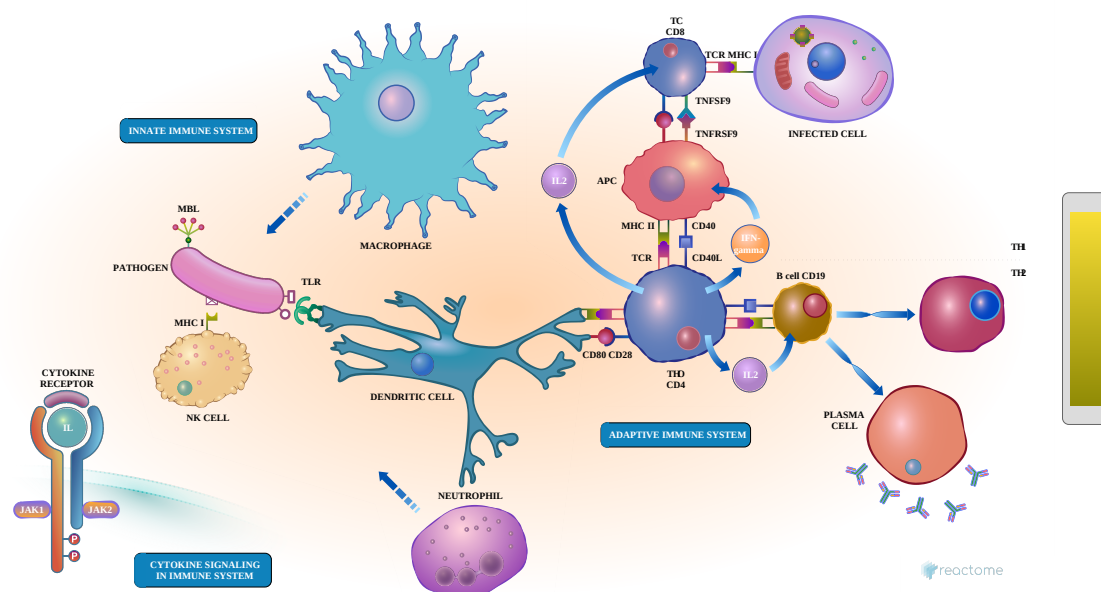
64 submitted entities found in this pathway, mapping to 101 Reactome entities

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ACTB	P60709, P63261	BTRC	Q9Y297	CFL1	P23528
CNTFR	P26992	CSF2	P04141	IFNA1	P01563, P01569, P01570, P0DY56
IFNA2	P01563	IFNAR1	P17181	IFNAR2	P48551-2

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IFNB1	P01574	IFNG	P01579	IFNGR1	P15260, P17181
IFNL1	Q8IU54	IFNL2	Q8IZI9, Q8IZJ0	IFNL3	Q8IZI9, Q8IZJ0
IFNLR1	Q13651, Q8IU57, Q969J5	IL10	P22301, Q9NPH9	IL10RA	Q13651, Q8IU57
IL10RB	P38484, Q08334	IL11RA	P08887, Q14626	IL12A	P29459
IL12B	P29460	IL12RB1	P42701	IL12RB2	Q99665
IL13	P35225	IL13RA1	P78552	IL13RA2	Q14627
IL17A	Q16552	IL18	Q14116, Q14116-2	IL19	Q9UHD0
IL20	Q9NYY1	IL20RA	Q9UHF4	IL20RB	Q6UXL0
IL21R	Q9HBE5	IL22	Q9GZX6	IL22RA1	Q8N6P7
IL22RA2	Q8N6P7, Q969J5	IL23A	Q9NPF7	IL23R	Q5VWK5
IL24	Q13007	IL26	Q9NPH9	IL27RA	Q6UWB1
IL2RA	P01589	IL2RB	P14784	IL37	Q9NZH6
IL4	P05112	IL4R	P24394	IL5RA	Q01344
IL6	P05231	IL6R	P08887, P08887-2	IL6ST	P40189, P40189-1, P40189-2
IL9	P15248	IL9R	Q01113	IRF5	Q13568
JAK1	P23458	NOD2	Q9HC29	PRLR	P16471, Q5VWK5
RELA	Q04206	RIPK2	O43353	STAT1	P42224, P42224-1, P42224-2
STAT3	P40763	STAT4	Q14765	TYK2	P29597
VIM	P08670				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CFL1	ENSG00000172757	CSF2	ENSG00000164400	IFNG	ENSG00000111537
IL10	ENSG00000136634	IL12A	ENSG00000168811	IL12B	ENSG00000113302
IL17A	ENSG00000112115	IL18	ENSG00000150782	IL23A	ENSG00000110944
IL23R	ENSG00000162594	IL4R	ENSG00000077238	IL6	ENSG00000136244
IL6R	ENSG00000160712	IRF5	ENSG00000128604	STAT1	ENSG00000115415
VIM	ENSG00000026025				

8. Immune System (R-HSA-168256)



Humans are exposed to millions of potential pathogens daily, through contact, ingestion, and inhalation. Our ability to avoid infection depends on the adaptive immune system and during the first critical hours and days of exposure to a new pathogen, our innate immune system.

References

Edit history

Date	Action	Author
2005-11-12	Created	Gillespie ME
2006-03-30	Authored	Luo F, Ouwehand WH, Gillespie ME, de Bono B
2006-04-19	Reviewed	Zwaginga JJ, D'Eustachio P, Gay NJ, Gale M Jr
2026-03-16	Modified	Weiser JD

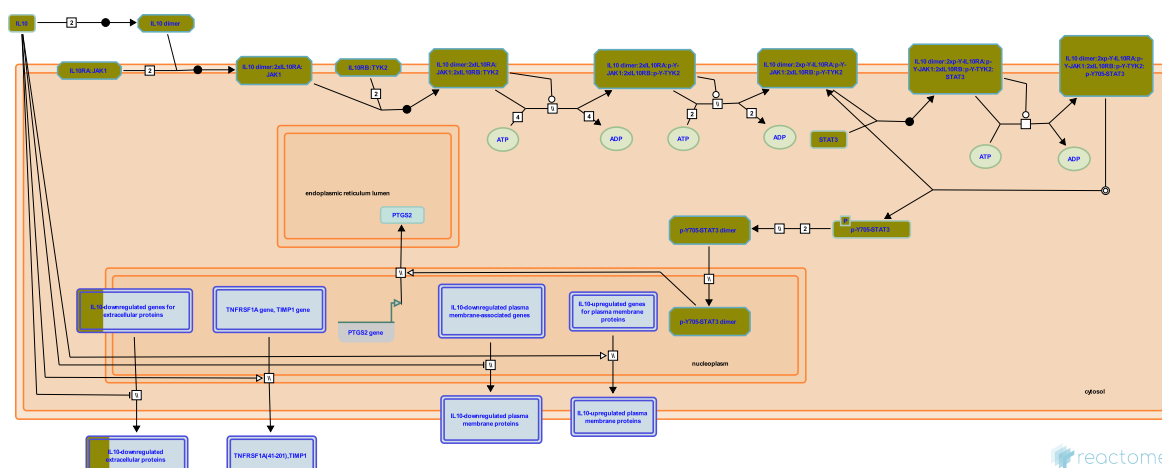
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ATP6V0C	P27449	BTRC	Q9Y297	CARD8	Q9C000
CFL1	P23528	CNTFR	P26992	CREBBP	Q92793
CSF2	P04141	CTLA4	P16410	GSN	P06396
IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563	IFNAR1	P17181
IFNAR2	P48551-2	IFNB1	P01574	IFNG	P01579
IFNGR1	P15260, P17181	IFNL1	Q8IU54	IFNL2	Q8IZI9, Q8IZJ0
IFNL3	Q8IZI9, Q8IZJ0	IFNLR1	Q13651, Q8IU57, Q969J5	IL10	P22301, Q9NPH9
IL10RA	Q13651, Q8IU57	IL10RB	P38484, Q08334	IL11RA	P08887, Q14626
IL12A	P29459	IL12B	P29460	IL12RB1	P42701
IL12RB2	Q99665	IL13	P35225	IL13RA1	P78552
IL13RA2	Q14627	IL17A	Q16552	IL18	Q14116, Q14116-2
IL19	Q9UHD0	IL20	Q9NYY1	IL20RA	Q9UHF4

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL20RB	Q6UXL0	IL21R	Q9HBE5	IL22	Q9GZX6
IL22RA1	Q8N6P7	IL22RA2	Q8N6P7, Q969J5	IL23A	Q9NPF7
IL23R	Q5VWK5	IL24	Q13007	IL26	Q9NPH9
IL27RA	Q6UWB1	IL2RA	P01589	IL2RB	P14784
IL37	Q9NZH6	IL4	P05112	IL4R	P24394
IL5RA	Q01344	IL6	P05231	IL6R	P08887, P08887-2
IL6ST	P40189, P40189-1, P40189-2	IL9	P15248	IL9R	Q01113
IRF5	Q13568	JAK1	P23458	LGALS3	P17931
NLRP3	Q96P20	NOD2	Q9HC29	OPTN	Q96CV9
PRLR	P16471, Q5VWK5	RELA	Q04206	RIPK2	O43353
STAT1	P42224, P42224-1, P42224-2	STAT3	P40763	STAT4	Q14765
TYK2	P29597	VCP	P55072	VIM	P08670

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IL10	ENSG00000136634	IL12A	ENSG00000168811	IL12B	ENSG00000113302
IL17A	ENSG00000112115	IL18	ENSG00000150782	IL23A	ENSG00000110944
IL23R	ENSG00000162594	IL4R	ENSG00000077238	IL6	ENSG00000136244
IL6R	ENSG00000160712	IRF5	ENSG00000128604	NLRP3	ENSG00000162711
STAT1	ENSG00000115415	VIM	ENSG00000026025		

9. Interleukin-10 signaling (R-HSA-6783783)



Interleukin-10 (IL10) was originally described as a factor named cytokine synthesis inhibitory factor that inhibited T-helper (Th) 1 activation and Th1 cytokine production (Fiorentino et al. 1989). It was found to be expressed by a variety of cell types including macrophages, dendritic cell subsets, B cells, several T-cell subpopulations including Th2 and T-regulatory cells (Tregs) and Natural Killer (NK) cells (Moore et al. 2001). It is now recognized that the biological effects of IL10 are directed at antigen-presenting cells (APCs) such as macrophages and dendritic cells (DCs), its effects on T-cell development and differentiation are largely indirect via inhibition of macrophage/dendritic cell activation and maturation (Pestka et al. 2004, Mocellin et al. 2004). T cells are thought to be the main source of IL10 (Hedrich & Bream 2010). IL10 inhibits a broad spectrum of activated macrophage/monocyte functions including monokine synthesis, NO production, and expression of class II MHC and costimulatory molecules such as IL12 and CD80/CD86 (de Waal Malefyt et al. 1991, Gazzinelli et al. 1992). Studies with recombinant cytokine and neutralizing antibodies revealed pleiotropic activities of IL10 on B, T, and mast cells (de Waal Malefyt et al. 1993, Rousset et al. 1992, Thompson-Snipes et al. 1991) and provided evidence for the *in vivo* significance of IL10 activities (Ishida et al. 1992, 1993). IL10 antagonizes the expression of MHC class II and the co-stimulatory molecules CD80/CD86 as well as the pro-inflammatory cytokines IL1β, IL6, IL8, TNFα and especially IL12 (Fiorentino et al. 1991, D'Andrea et al. 1993). The biological role of IL10 is not limited to inactivation of APCs, it also enhances B cell, granulocyte, mast cell, and keratinocyte growth/differentiation, as well as NK-cell and CD8⁺ cytotoxic T-cell activation (Moore et al. 2001, Hedrich & Bream 2010). IL10 also enhances NK-cell proliferation and/or production of IFN-γ (Cai et al. 1999).

IL10-deficient mice exhibited inflammatory bowel disease (IBD) and other exaggerated inflammatory responses (Kuhn et al. 1993, Berg et al. 1995) indicating a critical role for IL10 in limiting inflammatory responses. Dysregulation of IL10 is linked with susceptibility to numerous infectious and autoimmune diseases in humans and mouse models (Hedrich & Bream 2010).

IL10 signaling is initiated by binding of homodimeric IL10 to the extracellular domains of two adjoining IL10RA molecules. This tetramer then binds two IL10RB chains. IL10RB cannot bind to IL10 unless bound to IL10RA (Ding et al. 2001, Yoon et al. 2006); binding of IL10 to IL10RA without the co-presence of IL10RB fails to initiate signal transduction (Kotenko et al. 1997).

IL10 binding activates the receptor-associated Janus tyrosine kinases, JAK1 and TYK2, which are constitutively bound to IL10R1 and IL10R2 respectively. In the classic model of receptor activation assembly of the receptor complex is believed to enable JAK1/TYK2 to phosphorylate and activate each other. Alternatively the binding of IL10 may cause conformational changes that allow the pseudokinase inhibitory domain of one JAK kinase to move away from the kinase domain of the other JAK within the receptor dimer-JAK complex, allowing the two kinase domains to interact and trans-activate (Waters & Brooks 2015).

The activated JAK kinases phosphorylate the intracellular domains of the IL10R1 chains on specific tyrosine residues. These phosphorylated tyrosine residues and their flanking peptide sequences serve as temporary docking sites for the latent, cytosolic, transcription factor, STAT3. STAT3 transiently docks on the IL10R1 chain via its SH2 domain, and is in turn tyrosine phosphorylated by the receptor-associated JAKs. Once activated, it dissociates from the receptor, dimerizes with other STAT3 molecules, and translocates to the nucleus where it binds with high affinity to STAT-binding elements (SBEs) in the promoters of IL-10-inducible genes (Donnelly et al. 1999).

References

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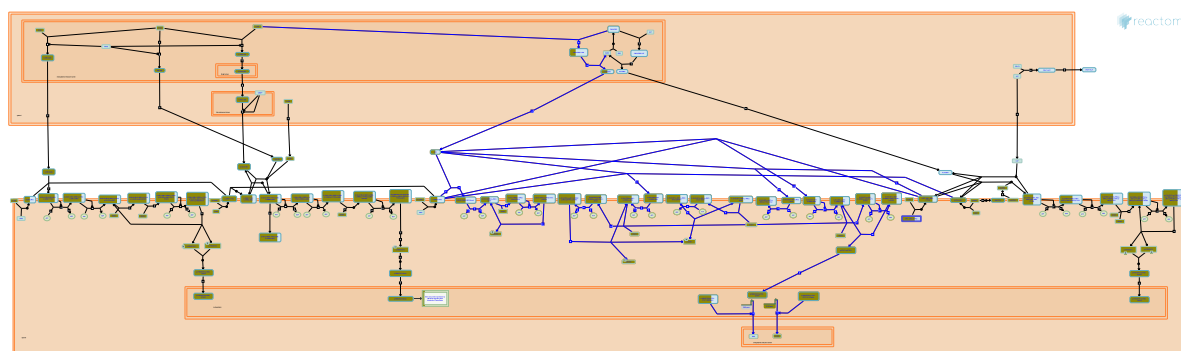
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2015-06-17	Authored	Jupe S
2015-06-17	Created	Jupe S
2016-09-05	Reviewed	Meldal BH
2016-11-14	Edited	Jupe S

12 submitted entities found in this pathway, mapping to 17 Reactome entities

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CSF2	P04141	IFNLR1	Q13651	IL10	P22301
IL10RA	Q13651	IL10RB	Q08334	IL12A	P29459
IL12B	P29460	IL18	Q14116	IL6	P05231
JAK1	P23458	STAT3	P40763	TYK2	P29597

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CSF2	ENSG00000164400	IL12A	ENSG00000168811	IL12B	ENSG00000113302
IL18	ENSG00000150782	IL6	ENSG00000136244		

10. Interleukin-35 Signalling (R-HSA-8984722)



Interleukin 35 (IL35) is an IL12 family cytokine produced by regulatory but not effector T-cells. It is a dimeric protein composed of IL-12RB2 and IL27RA chains. IL35 suppresses inflammatory responses of immune cells.

References

Egwuagu CE, Yu CR, Sun L & Wang R (2015). Interleukin 35: Critical regulator of immunity and lymphocyte-mediated diseases. *Cytokine Growth Factor Rev.*, 26, 587-93. [🔗](#)

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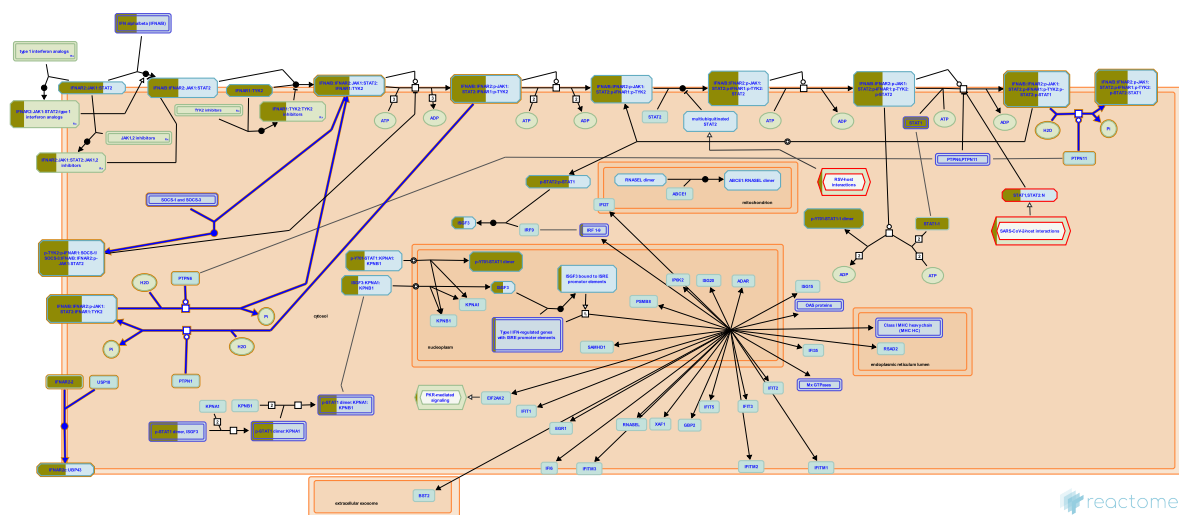
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2009-11-25	Authored	Jupe S
2015-11-09	Reviewed	Narazaki M, Tanaka M
2017-04-10	Created	Varusai TM
2017-05-11	Edited	Jupe S

9 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL12A	P29459	IL12RB2	Q99665	IL27RA	Q6UWB1
IL6ST	P40189	JAK1	P23458	STAT1	P42224
STAT3	P40763	STAT4	Q14765	TYK2	P29597

Input	Ensembl Id
IL12A	ENSG00000168811

11. Regulation of IFNA/IFNB signaling (R-HSA-912694)



There are several proteins and mechanisms involved in controlling the extent of ligand stimulation of IFNA/B signaling. These mechanisms can effect every step of the IFNA/B cascade. Dephosphorylation of JAK and STAT by SHP protein phosphatases, inhibition of STAT function in the nucleus by protein inhibitors of activated STATs (PIAS) proteins, inhibition of tyrosine kinase activity of JAKs by SOCS as well as inhibition of JAK and IFNAR2 interaction by UBP43 are few of the negative regulation mechanisms in controlling type I IFN signaling.

References

Fenner JE, Starr R, Cornish AL, Zhang JG, Metcalf D, Schreiber RD, ... Hertzog PJ (2006). Suppressor of cytokine signaling 1 regulates the immune response to infection by a unique inhibition of type I interferon activity. *Nat Immunol*, 7, 33-9. [↗](#)

Malakhova OA, Kim KI, Luo JK, Zou W, Kumar KG, Fuchs SY, ... Zhang DE (2006). UBP43 is a novel regulator of interferon signaling independent of its ISG15 isopeptidase activity. *EMBO J*, 25, 2358-67. [↗](#)

Hilton DJ (1999). Negative regulators of cytokine signal transduction. *Cell Mol Life Sci*, 55, 1568-77. [↗](#)

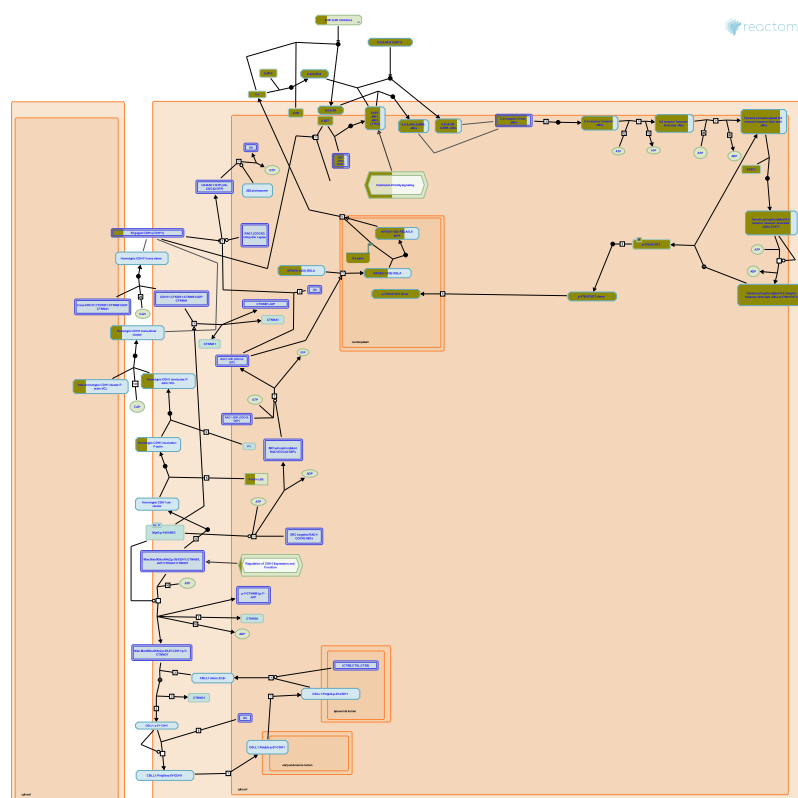
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2010-07-07	Edited	Garapati P V
2010-07-07	Authored	Garapati P V
2010-07-12	Created	Garapati P V
2010-08-17	Reviewed	Abdul-Sater AA, Schindler C

9 submitted entities found in this pathway, mapping to 13 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563	IFNAR1	P17181
IFNAR2	P48551-2	IFNB1	P01574	IFNGR1	P17181
JAK1	P23458	STAT1	P42224-1, P42224-2	TYK2	P29597

12. Activation of STAT3 by cadherin engagement (R-HSA-9958825)



The signal transducer and activator of transcription-3 (STAT3):

STAT3 is a latent cytoplasmic transcription factor, activated by receptor tyrosine kinases such as EGFR, or PDGFR, cytokine receptors such as the Interleukin-6 receptor family (IL6R or gp130 – also known as IL6ST), and non-receptor tyrosine kinases such as SRC. Activation entails phosphorylation of the critical tyrosine-705 residue (p-Y705) of STAT3 by the receptor itself or an associated Jak kinase, dimerization and translocation to the nucleus. The dimers subsequently bind specific DNA sequences to activate the transcription of specific genes involved in cell division and survival, while they downregulate the tumor suppressor p53 (Niu et al. 2005), thus protecting tumor cells from apoptosis (reviewed in Yu et al. 2009, Bharadwaj et al. 2020). Hyperactivation of STAT3 is present in a large number of cancers and has been reported to be required for tumour cell growth, survival, angiogenesis, metastasis and immune evasion (reviewed in Bharadwaj et al. 2020), while a hyperactive form of Stat3 (Stat3C) is able to transform cultured cells (Bromberg et al. 1999).

<u>Activation of Stat3 by cadherin engagement</u>

Early results demonstrated that cell to cell contact, as occurs with confluence, or aggregation of cultured cells, leads to a dramatic increase in phosphorylation of STAT3 at the activating tyrosine-705 (p-Y705-STAT3), DNA binding and transcriptional activity, in both non-transformed mouse or rat fibroblasts or MCF-10A human breast cells, as well as in a series of four human breast cancer cell lines (Vultur et al. 2004), and a number of human lung cancer cell lines (Geletu et al. 2012).

It was later definitively demonstrated that STAT3 is activated by direct homophilic interactions of E-cadherin (CDH1) molecules of the opposing cells, even in the absence of direct cell to cell contact. This was shown by plating mouse epithelial HC11 cells onto surfaces coated with fragments encompassing the two outermost domains of CDH1. In fact, CDH1 engagement was shown to trigger a dramatic surge in total RAC1 and CDC42 protein levels and activity through inhibition of proteolytic degradation, and this was responsible for the STAT3 activation observed. It was further demonstrated that the increase in RAC1/Cdc42 activity leads to transcriptional upregulation of members of the IL6 family of cytokines through the transcription factor NF- κ B and activation of the common receptor, gp130 (IL6ST), leading to STAT3 activation, in an autocrine manner. STAT3 then dimerizes, migrates to the nucleus and activates transcription of genes involved in cell division, survival, proliferation and migration (Arulanandam et al. 2009, Arulanandam et al. 2010, reviewed in Raptis et al. 2009, Raptis et al. 2011). In sharp contrast to STAT3, the activity of Erk1/2 (extracellular signal activated kinase, officially MAPK1/MAPK3) was unaffected by cell density, or cadherin engagement. Two mesenchymal cadherins, cadherin-11 (CDH11) and N-cadherin (CDH2) were also found to activate STAT3 in different cells, by similar mechanisms (Geletu, Arulanandam et al. 2013, reviewed in Geletu, Guy et al. 2013).

References

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- Bromberg JF, Wrzeszczynska MH, Devgan G, Zhao Y, Pestell RG, Albanese C & Darnell JE (1999). Stat3 as an oncogene. *Cell*, 98, 295-303. [🔗](#)
- Vultur A, Cao J, Arulanandam R, Turkson J, Jove R, Greer P, ... Raptis L (2004). Cell-to-cell adhesion modulates Stat3 activity in normal and breast carcinoma cells. *Oncogene*, 23, 2600-16. [🔗](#)

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Date	Action	Author
2025-05-25	Authored	Raptis L, Adan H, Daniel J
2025-06-25	Created	Orlic-Milacic M
2025-11-21	Reviewed	Raptis L
2026-03-03	Edited	Orlic-Milacic M
2026-03-16	Modified	Weiser JD

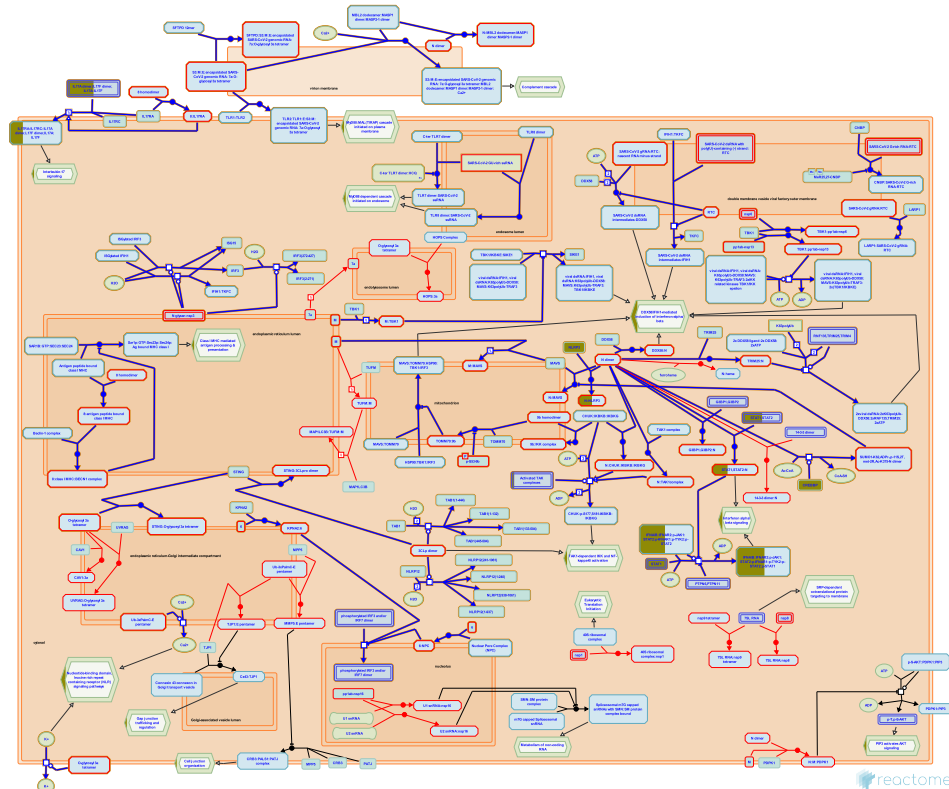
10 submitted entities found in this pathway, mapping to 15 Reactome entities

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ACTB	P60709, P63261	ACTC1	P68032	IL11RA	P08887
IL6	P05231	IL6R	P08887, P08887-2	IL6ST	P40189, P40189-1, P40189-2
JAK1	P23458	RELA	Q04206	STAT3	P40763

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TYK2	P29597				

Input	Ensembl Id
IL6	ENSG00000136244

13. SARS-CoV-2 activates/modulates innate and adaptive immune responses (R-HSA-9705671)



Diseases: COVID-19.

Coronaviruses (CoVs) are positive-sense RNA viruses that replicate in the interior of double membrane vesicles (DMV) in the cytoplasm of infected cells (Stertz S et al. 2007; Knoops K et al. 2008; V'kovski P et al. 2021). The viral replication and transcription are facilitated by virus-encoded non-structural proteins (SARS-CoV-2 nsp1–nsp16) that assemble to form a DMV-bound replication-transcription complex (RTC) (V'kovski P et al. 2021). The replication strategy of CoVs can generate both single-stranded RNA (ssRNA) and double-stranded RNA (dsRNA) species, that may act as pathogen-associated molecular patterns (PAMPs) recognized by pattern recognition receptor (PRR) such as toll-like receptor 7 (TLR7) and TLR8, antiviral innate immune response receptor RIG-I (also known as DEAD box protein 58, DDX58) and interferon-induced helicase C domain-containing protein 1 (IFIH1, also known as MDA5) (Salvi V et al. 2021; Campbell GR et al. 2021; Rebendenne A et al. 2021). The activated PRRs trigger signaling pathways to produce type I and type III interferons IFNs and proinflammatory mediators that perform antiviral functions. This Reactome module describes the mechanisms underlying PRR-mediated sensing of the severe acute respiratory syndrome coronavirus type 2 (SARS-CoV-2) infection. First, endosomal recognition of viral ssRNA occurs by means of TLR7 and TLR8, which detect GU-rich ssRNA sequences (Salvi V et al. 2021; Campbell GR et al. 2021). Second, SARS-CoV-2 dsRNA replication intermediates can be recognized by cytoplasmic receptors DDX58 and IFIH1 which bind to mitochondrial antiviral-signaling protein (MAVS, IPS-1) to induce the IFN-mediated antiviral response (Rebendenne A et al. 2021; Yin X et al. 2021). In addition, SARS-CoV-2 E can be sensed by TLR2 (Zheng M et al. 2021). Further, cellular nucleic acid-binding protein (CNBP) and La-related protein 1 (LARP1) can directly bind SARS-CoV-2 gRNA to repress SARS-CoV-2 replication (Schmidt N et al. 2021). This module also describes several strategies developed by SARS-CoV-2 to evade or alter host immunity, including escaping innate immune

sensors, inhibiting IFN production and signaling, and evading antiviral function of IFN stimulated gene (ISG) products. For example, SARS-CoV-2 encodes nsp14 and nsp16 which possess guanine-N7-methyltransferase activity and 2'-O-methyl-transferase activity respectively (Ogando NS et al. 2020; Krafcikova P et al. 2020; Viswanatha T et al. 2020; Lin S et al. 2021; Yan L et al. 2021). In human coronaviruses nsp14 generates 5' cap-0 viral RNA (m7GpppN, guanine N7-methylated) and nsp16 further methylates cap-0 viral RNA. These viral RNA modifications mimic the 5'-cap structure of host mRNAs allowing the virus to efficiently evade recognition by cytosolic DDX58 and IFIH1 (Chen Y et al. 2009, 2011; Daffis S et al. 2010, shown for CoVs such as SARS-CoV-1 and MERS-CoV). Structural studies and computational analysis suggest that properties and biological functions of SARS-CoV-2 nsp14 and nsp16 could be very similar to these of SARS-CoV-1 (Rosas-Lemus M et al. 2020; Lin S et al. 2020; Viswanathan T et al. 2020; Krafcikova P et al. 2020; Jiang Y et al. 2020; Wilamowski M et al. 2021). Further, the uridylylate-specific endoribonuclease (EndoU) activity of SARS-CoV-2 nsp15 degrades viral RNA to hide it from innate immune sensors (Frazier MN et al. 2021). Moreover, SARS-CoV-2 encodes several proteins that directly bind to host targets associated with SARS-CoV-2 infection and cytokine production (Shin D et al. 2020; Viswanathan T et al. 2020; Xia H et al. 2020; Matsuyama T et al. 2020; Yuen CK et al. 2020; reviewed by Park A & Iwasaki A 2020). This Reactome module describes several such binding events and their consequences. For example, as a deubiquitinating and deISGylating enzyme, viral nsp3 binds to and removes ISG15 from signaling proteins such as IRF3 and IFIH1 thereby modulating the formation of signaling complexes and the activation of IRF3/7 and NF-kappaB (Liu CQ et al. 2021). Binding of SARS-CoV-2 nsp6, nsp13 or membrane (M) protein to cytosolic TBK1 prevents IRF3/7 activation and inhibits IFN production downstream of DDX58, IFIH1, MAVS and STING signaling pathways (Xia H et al. 2020; Sui L et al. 2021). Next, M protein targets MAVS to prevent the formation of the MAVS signalosome complex and thereby inhibits downstream signaling pathways of DDX58 and IFIH1 (Fu YZ et al. 2021). Binding of SARS-CoV-2 nucleocapsid (N) protein to E3 ubiquitin ligase TRIM25 inhibits TRIM25-mediated DDX58 ubiquitination and the DDX58 signaling pathway (Gori SG et al. 2021). N interacts with NLRP3 to promote the assembly and activation of the NLRP3 inflammasome (Pan P et al. 2021). The interaction between viral N and MASP2 promotes MASP2-mediated cleavage of C4 (Ali YM et al. 2021) and C2 (Kang S et al. 2021) leading to the hyperactivation of the complement system. Besides, viral N promotes NF-kappaB activation by targeting signaling complexes of TAK1 and IKK (Wu Y et al. 2021). The ion channel activities of accessory protein ORF3a or 3a (open reading frame 3a) and SARS-CoV-2 envelope (E) protein contribute to activation of the NLRP3 inflammasome leading to highly inflammatory pyroptotic cell death (based on findings for SARS-CoV-1, Siu KL et al. 2019). SARS-CoV-2 nsp5 protease (3CLpro) cleaves TAB1, a component of the TAK1 complex, thus inhibiting NF-kappaB activation (Moustaqil M et al. 2021). 3CLpro targets NLRP12 which modulates the expression of inflammatory cytokines through the regulation of the NFkappaB and MAPK pathways (Moustaqil M et al. 2021). SARS-CoV-2 6 (ORF6) interacts with importin KPNA2 and components of the nuclear pore complex, NUP98 and RAE1, to block nuclear translocation of IRF3, STAT1 and STAT2 (Xia H et al. 2020; Miorin L et al. 2020). SARS-CoV-2 9b (ORF9b) inhibits the MAVS-mediated production of type I IFNs by targeting TOMM70 on the mitochondria (Jiang HW et al. 2020). Binding of mitochondrial viral 9 to IKBKG prevents MAVS-dependent NF-kappaB activation (Wu J et al. 2021). Although the evasion mechanisms are mainly conserved between SARS-CoV-1 and SARS-CoV-2 (Gordon DE et al. 2020), studies identified SARS-CoV-2-specific modulations of host immune response that may contribute to pathophysiological determinants of COVID-19 (Gordon DE et al. 2020; Schiller HB et al. 2021). This Reactome module describes several virus-host interactions identified in cells during SARS-CoV-2, but not SARS-CoV-1, infection. For example, SARS-CoV-2 8 (ORF8) regulates the expression of class I MHC on the surface of the infected cells through

an autophagy-dependent lysosomal degradation of class I MHC (Zhang Y et al. 2021). At the plasma membrane, binding of secreted viral 8 to IL17RA activates IL17 signaling pathway leading to an increased secretion of cytokines/chemokines thus contributing to cytokine storm during SARS-CoV-2 infection (Lin X et al. 2021). Furthermore, SARS-CoV-2-host interactome and proteomics studies identified various human proteins that are targeted by SARS-CoV-2 proteins (Gordon DE et al. 2020a, b; Bojkova D et al. 2020; Stukalov A et al. 2021; Li J et al. 2021; Messina F et al. 2021). This Reactome module does not cover all identified SARS-CoV-2-human interactions; the module describes those associations that were functionally validated.

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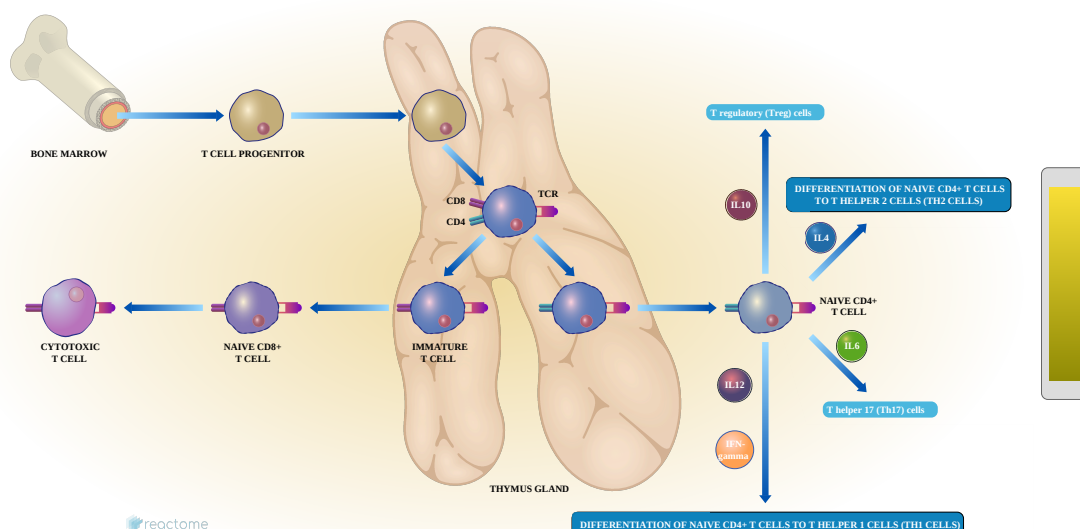
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2021-10-27	Authored	Shamovsky V
2022-02-18	Edited	Shamovsky V
2022-02-18	Reviewed	Messina F
2025-11-07	Modified	Shamovsky V

14 submitted entities found in this pathway, mapping to 19 Reactome entities

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CREBBP	Q92793	IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563
IFNAR1	P17181	IFNAR2	P48551-2	IFNB1	P01574
IFNGR1	P17181	IL17A	Q16552	JAK1	P23458
NLRP3	Q96P20	NOD2	Q9HC29	RIPK2	O43353
STAT1	P42224, P42224-1, P42224-2	TYK2	P29597		

14. Differentiation of T cells ([R-HSA-9945266](#))



Thymic seeding progenitors (TSPs), originating from the bone marrow, enter the thymus and pass through six double negative (CD4-/CD8-) stages: DN1, DN2a, DN2b, DN3a, DN3b, and DN4 (reviewed in Sun et al. 2023). T cell receptor components are initially expressed in DN3b cells. Double negative 4 (DN4) cells then express both CD4 and CD8 and are designated double positive (DP) cells. DP cells then differentiate into single positive cells: naive (not yet activated by antigen) CD4+ T cells and naive CD8+ T cells.

Upon stimulation of the T cell receptor (TCR) by antigen, CD4+ T cells are directed by other external factors, notably cytokines, to differentiate into various types of T helper cells. The different types of T helper cells are characterized by master regulatory transcription factors and secreted effector cytokines (reviewed in Lee 2023, Sun et al. 2023):

T helper 1 (Th1) cells are regulated by TBX21 (T-bet) and secrete interferon gamma (IFNG).

T helper 2 (Th2) cells are regulated by GATA3 and secrete IL4, IL5, and IL13.

T helper 9 (Th9) cells are regulated by IRF4 and SPI1 (PU.1) and secrete IL9, IL10, and IL21.

T helper 17 (Th17) cells are regulated by RORC isoform t (RORgammat) and STAT3 (reviewed in Imam et al. 2018) and secrete IL17A, IL17F, and IL22.

T helper 22 (Th22) cells are regulated by TBX21 and STAT3 and secrete IL22, but not IFNG or IL17 (reviewed in Imam et al. 2018).

T follicular helper (Tfh) cells are regulated by BCL6 and secrete IL21.

T regulatory (Treg) cells are regulated by FOXP3 and secrete IL10.

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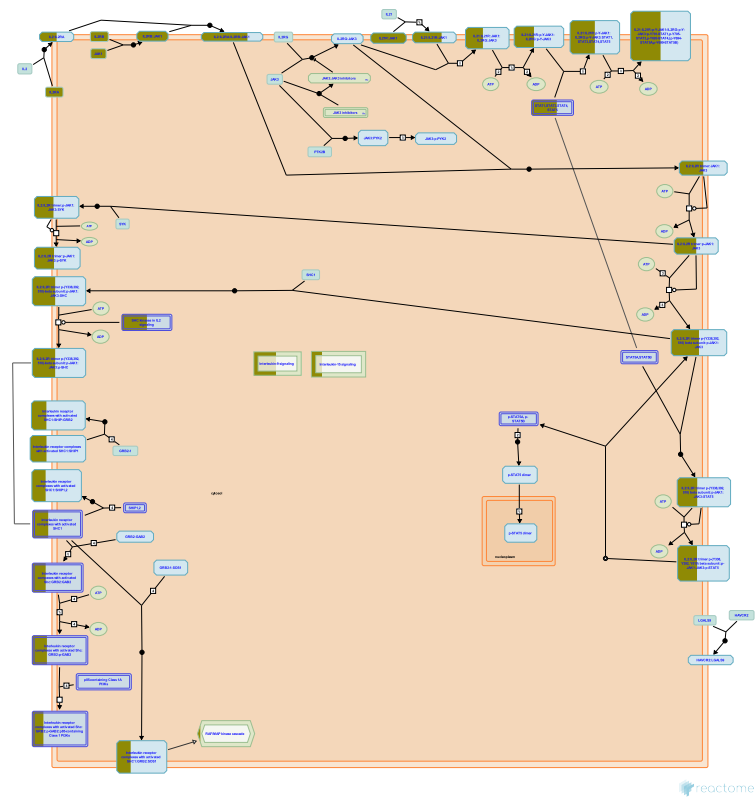
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2025-04-12	Created	May B
2025-08-04	Reviewed	Barbutti I
2025-08-05	Reviewed	Lord GM
2026-02-24	Reviewed	Barbutti I
2026-02-26	Reviewed	Lord GM
2026-02-28	Modified	May B

9 submitted entities found in this pathway, mapping to 14 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CREBBP	Q92793	IFNG	P01579	IL12RB2	Q99665
IL13	P35225	IL4	P05112	IL4R	P24394
RNF2	Q99496	STAT1	P42224-1	STAT4	Q14765

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IFNG	ENSG00000111537	IL12RB2	ENSG00000081985	IL13	ENSG00000169194
IL4	ENSG00000113520	IL4R	ENSG00000077238		

15. Interleukin-2 family signaling (R-HSA-451927)



Cellular compartments: plasma membrane.

The interleukin-2 family (also called the common gamma chain cytokine family) consists of interleukin (IL)2, IL9, IL15 and IL21. Although sometimes considered to be within this family, the IL4 and IL7 receptors can form complexes with other receptor chains and are represented separately in Reactome. Receptors of this family associate with JAK1 and JAK3, primarily activating STAT5, although certain family members can also activate STAT1, STAT3 or STAT6.

References

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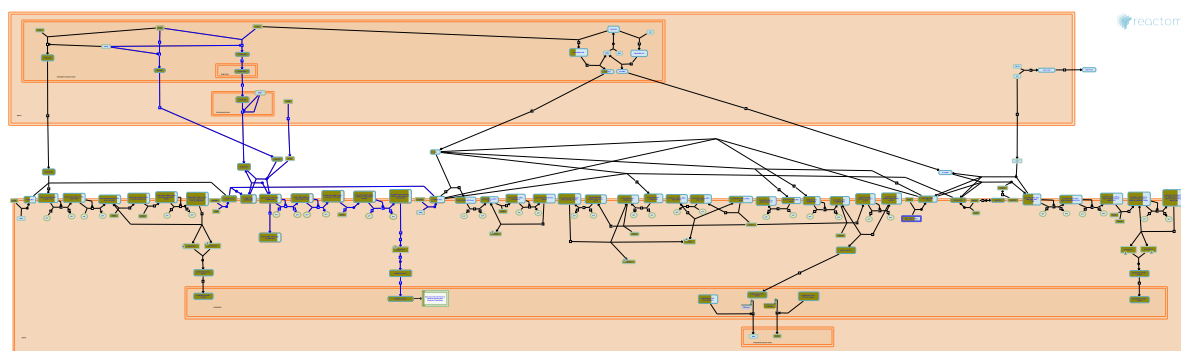
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Date	Action	Author
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2010-05-17	Authored	Ray KP
2010-08-06	Edited	Jupe S
2011-02-11	Reviewed	Villarino A
2011-03-17	Reviewed	Dooms H

11 submitted entities found in this pathway, mapping to 11 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CSF2	P04141	IL21R	Q9HBE5	IL2RA	P01589
IL2RB	P14784	IL5RA	Q01344	IL9	P15248
IL9R	Q01113	JAK1	P23458	STAT1	P42224
STAT3	P40763	STAT4	Q14765		

16. Interleukin-12 signaling (R-HSA-9020591)



Interleukin 12 (IL-12) is heterodimeric cytokine produced by dendritic cells, macrophages and neutrophils. It is encoded by the genes Interleukin-12 subunit alpha (IL12A) and Interleukin-12 subunit beta (IL12B), which encode a 35-kDa light chain (p35) and a 40-kDa heavy chain (p40), respectively. The active IL12 heterodimer is sometimes referred to as p70. The p35 component has homology to single-chain cytokines, while p40 is homologous to the extracellular domains of members of the haematopoietic cytokine-receptor family. The IL12 heterodimer therefore resembles a cytokine linked to a soluble receptor.

IL12 is involved in the differentiation of naive T cells into Th1 cells and sometimes known as T cell-stimulating factor. IL12 enhances the cytotoxic activity of Natural Killer cells and CD8+ cytotoxic T lymphocytes. IL12 also has anti-angiogenic activity, mediated by increased production of CXCL10 via interferon gamma.

The IL12 receptor is a heterodimer formed by Interleukin-12 receptor subunit beta-1 (IL12RB1) and Interleukin-12 receptor subunit beta-2 (IL12RB2), both of which have extensive homology to IL6ST (gp130), the signal transducing receptor subunit of the IL6-like cytokine superfamily. IL-12RB2 is considered to play the key role in IL12 function, in part because its expression on activated T cells is stimulated by cytokines that promote Th1 cell development and inhibited by those that promote Th2 cells development. In addition, IL12 binding leads to IL12RB2 tyrosine phosphorylation, which provides binding sites for the kinases Non-receptor tyrosine-protein kinase TYK2 and Tyrosine-protein kinase JAK2. These activate transcription factor proteins in the Signal transducer and activator of transcription (STAT) family, particularly STAT4.

References

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Edit history

Date	Action	Author
2014-06-04	Authored	Jupe S
2016-01-28	Reviewed	Meldal BH
2017-05-12	Reviewed	van de Vosse E
2017-05-17	Authored	Duenas C
2017-09-07	Created	Duenas C

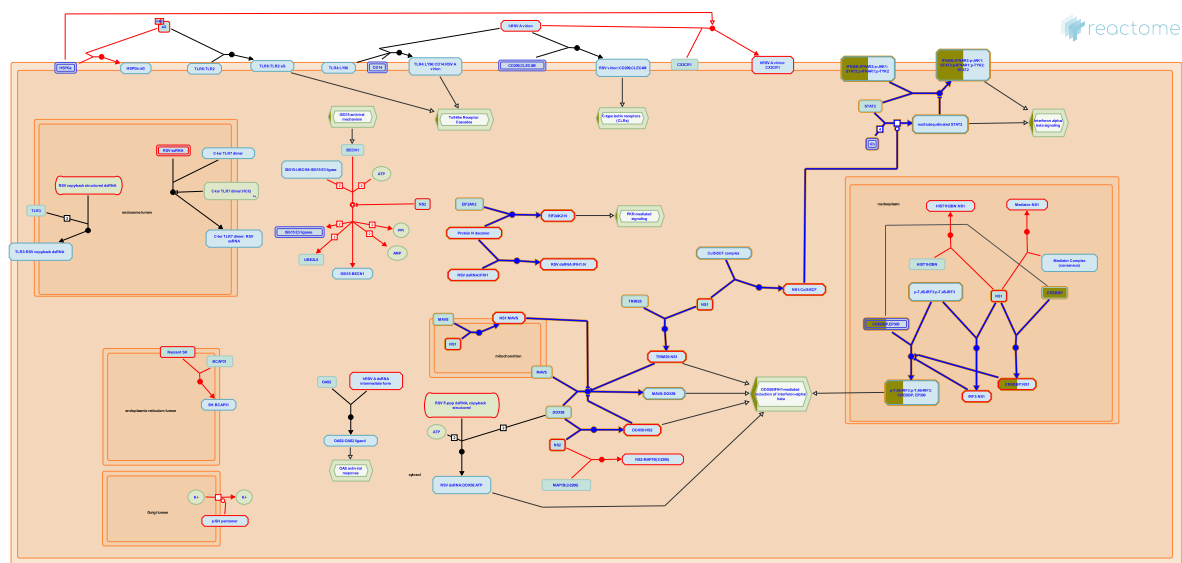
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2017-11-15	Edited	Jupe S
2026-03-16	Modified	Weiser JD

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Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CFL1	P23528	IFNG	P01579	IL10	P22301
IL12A	P29459	IL12B	P29460	IL12RB1	P42701
IL12RB2	Q99665	JAK1	P23458	STAT4	Q14765
TYK2	P29597				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CFL1	ENSG00000172757	IFNG	ENSG00000111537	IL10	ENSG00000136634

17. Evasion by RSV of host interferon responses (R-HSA-9833109)



Cellular compartments: cytosol.

Diseases: respiratory syncytial virus infectious disease.

Infection with human respiratory syncytial virus (hRSV) is typically associated with low to undetectable levels of type I interferons (IFNs). Several hRSV proteins interact with host innate immune system factors that support the type I interferon response. Nonstructural proteins NS1 and NS2 localize to mitochondria and nucleus where they bind MAVS, DDX58, TRIM25, IRF3, and CREBBP, affecting DDX58/IFIH1-mediated interferon induction (reviewed by Thornhill & Verhoeven, 2020). Additionally, hRSV nucleoprotein interacts with MDA5 downregulating the interferon response and with PKR (EIF2AK2) blocking the innate immune system signal for shutting down protein translation. Further interactions of the M, SH, and G proteins are reviewed by van Royen et al, 2022.

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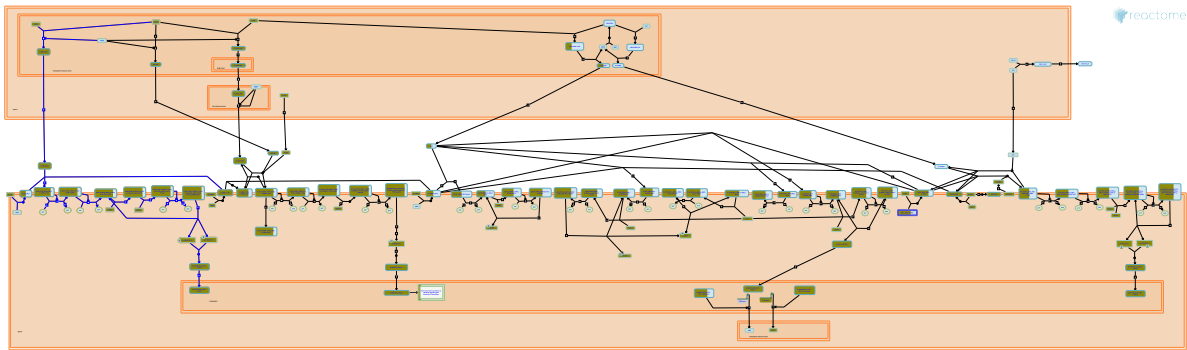
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2023-04-13	Created	Stephan R
2023-11-03	Reviewed	Bergeron HC
2023-11-09	Modified	Stephan R
2023-11-09	Edited	Stephan R

9 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CREBBP	Q92793	IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563
IFNAR1	P17181	IFNAR2	P48551-2	IFNB1	P01574
IFNGR1	P17181	JAK1	P23458	TYK2	P29597

18. Interleukin-23 signaling (R-HSA-9020933)



Interleukin-23 (IL23) is a heterodimer of Interleukin-12 subunit beta (IL12B, IL-12p40), which is shared with IL12, and Interleukin-23 subunit alpha IL23A (IL-23p19) subunit. The functional receptor for IL23 consists of Interleukin-12 receptor subunit beta-1 (IL12RB1), which is shared with the IL12 receptor, and Interleukin-23 receptor (IL23R). IL23R is mainly expressed on activated memory T cells, Natural Killer cells, monocytes/macrophage and at low levels on dendritic cells (DCs). IL23 is mainly secreted by activated macrophages and DCs in peripheral tissues such as skin, intestinal mucosa and lung.

IL23 is proinflammatory and implicated in several autoimmune inflammatory disorders such as colitis, gastritis, psoriasis and arthritis. It is similar to IL-12 both in structure and its ability to memory T cells to increase interferon- γ (IFN- γ) production and proliferation, the ability of IL-23 to induce IL-17.

IL23 activates the Janus kinases JAK2 and TYK2, resulting in phosphorylation of the receptor complex, which forms the docking sites for Signal transducer and activator of transcription 3 (STAT3) and STAT4 to bind and become phosphorylated.

References

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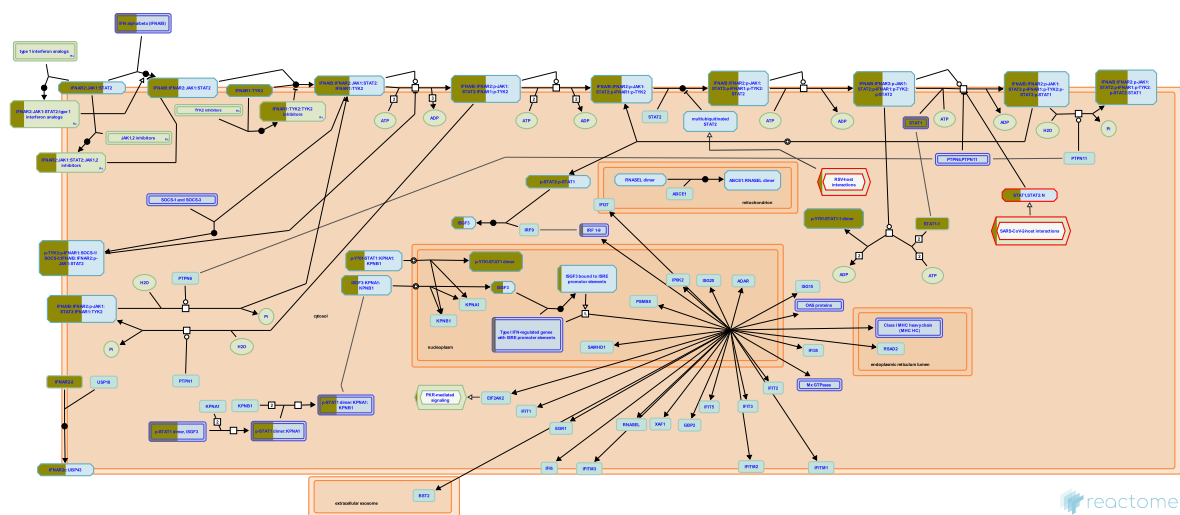
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2016-12-02	Authored	Duenas C
2017-05-12	Reviewed	van de Vosse E
2017-09-12	Created	Jupe S
2017-11-10	Edited	Jupe S

8 submitted entities found in this pathway, mapping to 8 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL12B	P29460	IL12RB1	P42701	IL23A	Q9NPF7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL23R	Q5VWK5	PRLR	Q5VWK5	STAT3	P40763
STAT4	Q14765	TYK2	P29597		

19. Interferon alpha/beta signaling (R-HSA-909733)



Type I interferons (IFNs) are composed of various genes including IFN alpha (IFNA), beta (IFNB), omega, epsilon, and kappa. In humans the IFNA genes are composed of more than 13 subfamily genes, whereas there is only one IFNB gene. The large family of IFNA/B proteins all bind to a single receptor which is composed of two distinct chains: IFNAR1 and IFNAR2. The IFNA/B stimulation of the IFNA receptor complex leads to the formation of two transcriptional activator complexes: IFNA-activated-factor (AAF), which is a homodimer of STAT1 and IFN-stimulated gene factor 3 (ISGF3), which comprises STAT1, STAT2 and a member of the IRF family, IRF9/P48. AAF mediates activation of the IRF-1 gene by binding to GAS (IFNG-activated site), whereas ISGF3 activates several IFN-inducible genes including IRF3 and IRF7.

References

Uzé G, Schreiber G, Piehler J & Pellegrini S (2007). The receptor of the type I interferon family. *Curr Top Microbiol Immunol*, 316, 71-95. [↗](#)

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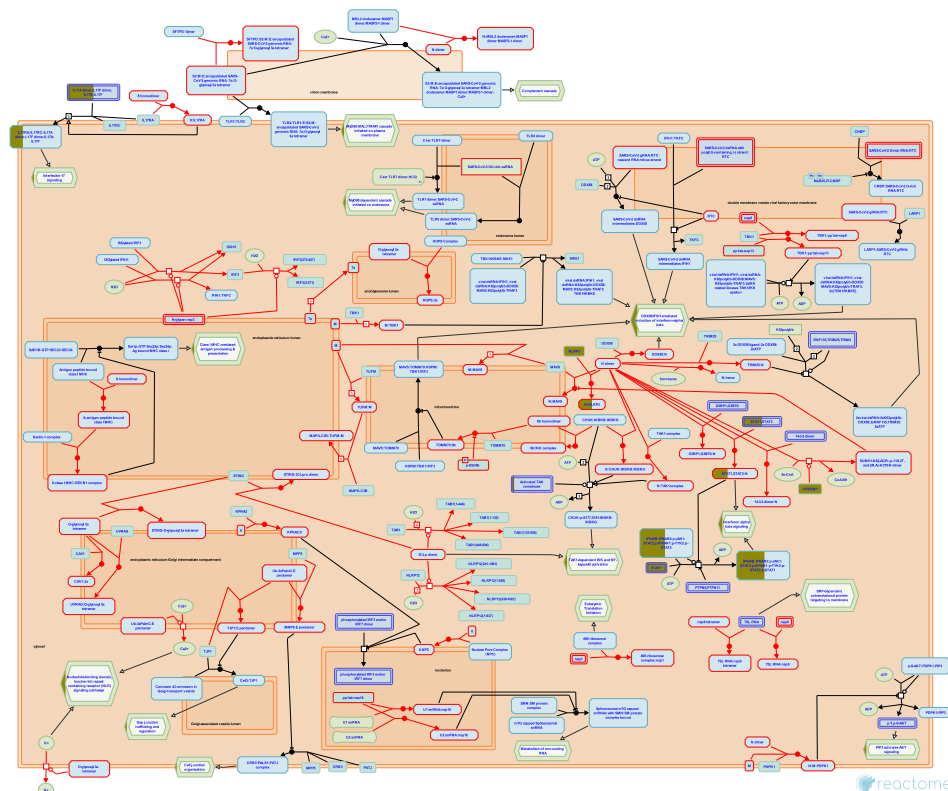
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2010-08-17	Reviewed	Abdul-Sater AA, Schindler C

10 submitted entities found in this pathway, mapping to 16 Reactome entities

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IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563	IFNAR1	P17181
IFNAR2	P48551-2	IFNB1	P01574	IFNGR1	P17181
IRF5	Q13568	JAK1	P23458	STAT1	P42224, P42224-1, P42224-2
TYK2	P29597				

Input	Ensembl Id
IRF5	ENSG00000128604

20. SARS-CoV-2-host interactions (R-HSA-9705683)



Diseases: COVID-19.

Coronaviruses are a group of enveloped viruses with single-stranded, positive-sense RNA genomes. Each of the steps of viral replication - attachment and entry, translation of viral replicase, genome transcription and replication, translation of structural proteins, and virion assembly and release - involves host factors. These interactions can cause alterations in cellular structure and physiology, and activate host stress responses, autophagy, cell death, and processes of innate immunity (Fung TS & Liu DX 2019). This Reactome module describes molecular mechanisms by which severe acute respiratory syndrome coronavirus type 2 (SARS-CoV-2) modulates innate and adaptive immune responses, autophagy, host translation, intracellular signaling and regulatory pathways, and PDZ-mediated cell-cell junctions, mostly annotated from studies of cells infected with SARS-CoV-2.

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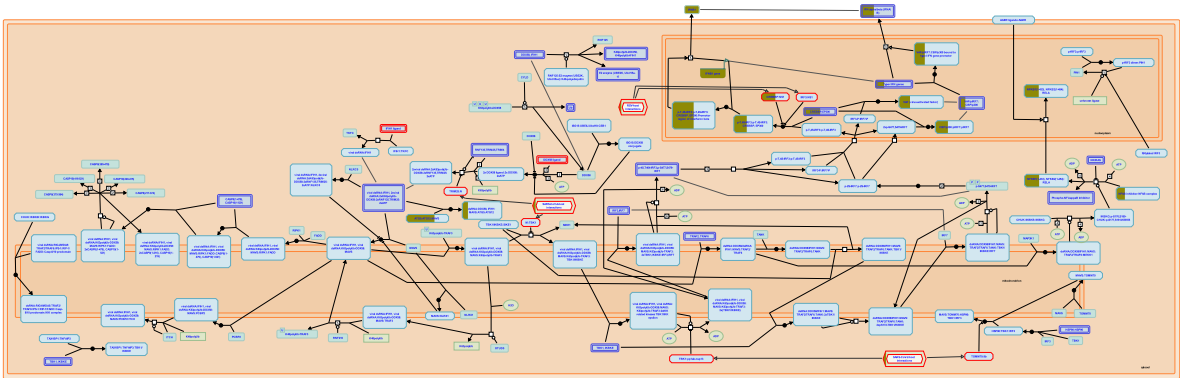
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2021-10-27	Authored	Shamovsky V
2022-02-18	Edited	Shamovsky V
2022-02-18	Reviewed	Messina F
2022-03-01	Reviewed	Gillespie ME
2023-10-12	Modified	Weiser JD

14 submitted entities found in this pathway, mapping to 19 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CREBBP	Q92793	IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563
IFNAR1	P17181	IFNAR2	P48551-2	IFNB1	P01574
IFNGR1	P17181	IL17A	Q16552	JAK1	P23458
NLRP3	Q96P20	NOD2	Q9HC29	RIPK2	O43353
STAT1	P42224, P42224-1, P42224-2	TYK2	P29597		

21. DDX58/IFIH1-mediated induction of interferon-alpha/beta (R-HSA-168928)



reactome

Cellular compartments: mitochondrial outer membrane.

RIG-I-like helicases (RLHs) the retinoic acid inducible gene-I (RIG-I, DDX58) and melanoma differentiation associated gene 5 (MDA5, IFIH1) are RNA helicases that recognize viral double-stranded RNA (dsRNA) present within the cytoplasm (Yoneyama M & Fujita T 2007, 2008). Upon viral infection dsRNA is generated by positive-strand RNA virus families such as Flaviviridae and Coronaviridae, negative-strand RNA virus families including Orthomyxoviridae and Paramyxoviridae, and DNA virus families such as Herpesviridae and Adenoviridae (Weber F et al. 2006; Son KN et al. 2015). Functionally RIG-I (DDX58) and MDA5 (IFIH1) positively regulate the IFN genes in a similar fashion, however they differ in their response to different viral species. DDX58 (RIG-I) is essential for detecting influenza virus, Sendai virus, VSV and Japanese encephalitis virus (JEV), whereas IFIH1 (MDA5) is essential in sensing encephalomyocarditis virus (EMCV), Mengo virus and Theiler's virus, all of which belong to the picornavirus family. RIG-I and MDA5 signalling results in the activation of IKK epsilon and (TBK binding kinase 1) TBK1, two serine/threonine kinases that phosphorylate interferon regulatory factor 3 and 7 (IRF3 and IRF7). Upon phosphorylation, IRF3 and IRF7 translocate to the nucleus and subsequently induce interferon alpha (IFNA) and interferon beta (IFNB) gene transcription (Yoneyama M et al. 2004; Yoneyama M & Fujita T 2007, 2008).

References

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Loo YM, Fornek J, Crochet N, Bajwa G, Perwitasari O, Martinez-Sobrido L, ... Gale M Jr (2008). Distinct RIG-I and MDA5 signaling by RNA viruses in innate immunity. *J Virol*, 82, 335-45. [🔗](#)

Bowie AG & Unterholzner L (2008). Viral evasion and subversion of pattern-recognition receptor signalling. *Nat Rev Immunol*, 8, 911-22. [🔗](#)

Yoneyama M & Fujita T (2008). Structural mechanism of RNA recognition by the RIG-I-like receptors. *Immunity*, 29, 178-81. [🔗](#)

Yoneyama M & Fujita T (2007). RIG-I family RNA helicases: cytoplasmic sensor for antiviral innate immunity. *Cytokine Growth Factor Rev*, 18, 545-51. [🔗](#)

Edit history

Date	Action	Author
2005-11-22	Created	de Bono B

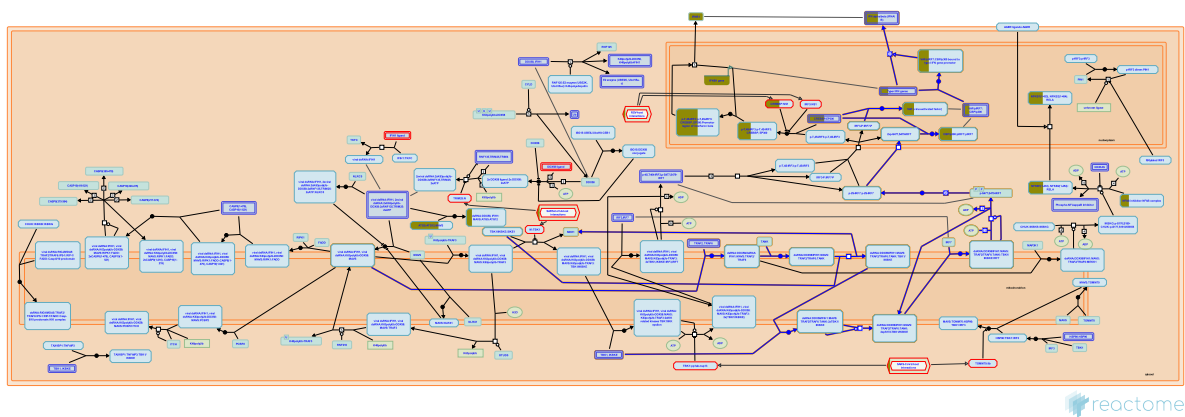
Date	Action	Author
2010-08-02	Edited	Garapati P V
2010-08-02	Authored	Garapati P V
2010-10-30	Reviewed	Akira S, Kawai T
2026-03-16	Modified	Weiser JD

8 submitted entities found in this pathway, mapping to 13 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ATG12	O94817	ATG5	Q9H1Y0	CREBBP	Q92793
IFNA1	P01563, P01569, P01570, P0DY56	IFNA2	P01563	IFNB1	P01574
RELA	Q04206				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IFNA13	ENSG00000233816	IFNA2	ENSG00000188379	IFNB1	ENSG00000171855

22. TRAF6 mediated IRF7 activation (R-HSA-933541)



Cellular compartments: mitochondrial outer membrane.

TRAF6 is crucial for both DDX58 (RIG-I)- and IFIH1 (MDA5)-mediated antiviral responses. The absence of TRAF6 resulted in enhanced viral replication and a significant reduction in the production of type I IFNs and IL6 after infection with RNA virus. Activation of NF- κ B and IRF7, but not that of IRF3, was significantly impaired during RIG-like helicases (RLHs) signaling in the absence of TRAF6. TRAF6-induced activation of IRF is likely to be specific for IRF7, while TRAF3 is thought to activate both IRF3 and IRF7. These results strongly suggest that the TRAF6- and TRAF3-dependent pathways are likely to bifurcate at mitochondrial antiviral-signaling protein MAVS (IPS-1), but to converge later at IRF7 in order to co-operatively induce sufficient production of type I IFNs during RLH signaling.

References

Konno H, Yamamoto T, Yamazaki K, Gohda J, Akiyama T, Semba K, ... Inoue J (2009). TRAF6 establishes innate immune responses by activating NF- κ B and IRF7 upon sensing cytosolic viral RNA and DNA. PLoS One, 4, e5674. [🔗](#)

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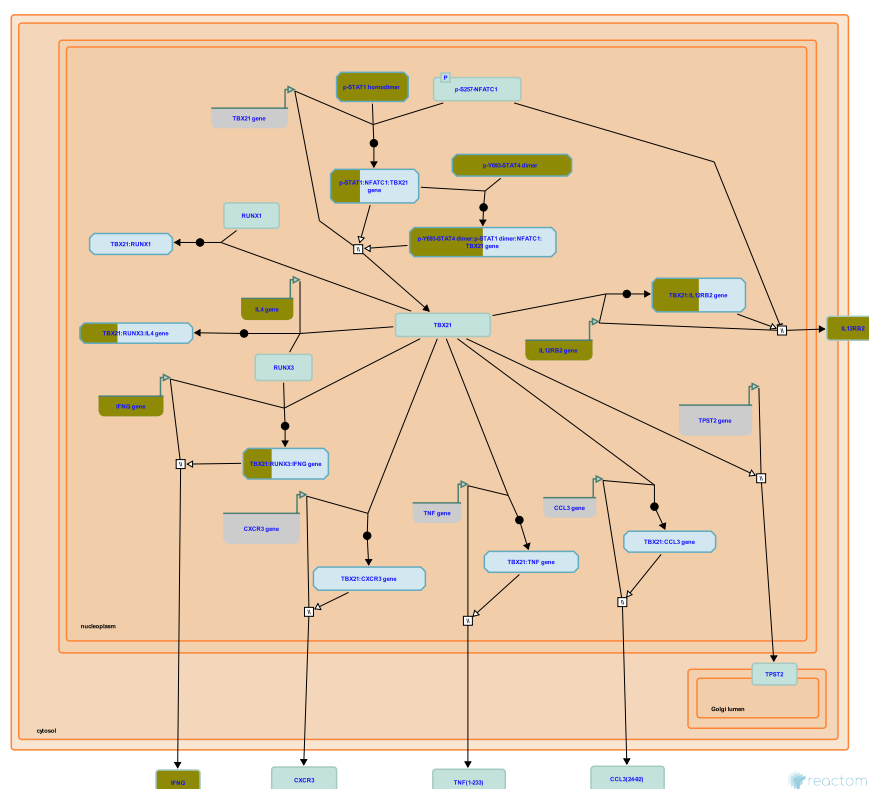
Date	Action	Author
2010-08-02	Edited	Garapati P V
2010-08-02	Authored	Garapati P V
2010-08-16	Created	Garapati P V
2010-10-30	Reviewed	Akira S, Kawai T
2026-03-16	Modified	Weiser JD

5 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id
CREBBP	Q92793	IFNA1	P01563, P01569, P01570, P0DY56
IFNA2	P01563	IFNB1	P01574

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IFNA13	ENSG00000233816	IFNA2	ENSG00000188379	IFNB1	ENSG00000171855

23. Differentiation of naive CD4+ T cells to T helper 1 cells (Th1 cells) ([R-HSA-9942503](#))



Cellular compartments: nucleoplasm.

A conjunction of signaling from the T cell receptor, Interferon gamma (IFNG), and Interleukin-12 (IL12) induces the differentiation of T helper 1 cells (Th1) by activating the master regulator TBX21 (T-bet). Expression of TBX21 (T-bet) occurs in two phases. The first phase is dependent on IFNG acting via STAT1 and the T cell receptor acting via calcium, calcineurin, and NFATC1 (inferred from mouse homologs in Zhong et al. 2025). During the first phase, TBX21 activates expression of IL21RB2, a subunit of the receptor for IL12 (inferred from mouse homologs in Mullen et al. 2001, Afkarian et al. 2002, Schulz et al. 2009, Zhu et al. 2012). The second phase is dependent on IL12 acting via IL21RB2 and STAT4 (Inferred from mouse cells in Schulz et al. 2009, Wei et al. 2010, Zhu et al. 2012, Fang et al. 2022).

TBX21 with RUNX3 then enhances expression of IFNG (inferred from mouse homologs in Szabo et al. 2000, Mullen et al. 2001, Hatton et al. 2006, Djuretic et al. 2007, Jenner et al. 2009, Zhu et al. 2012), thereby creating a positive feedback loop that maintains TBX21 expression. TBX21 also binds the transcription factor RUNX1 and prevents RUNX1 from activating expression of RORC, an activator of Th17 differentiation (inferred from mouse homologs in Lazarevic et al. 2011). TBX21 also represses expression of IL4, an activator of Th2 differentiation (Jenner et al. 2009, and inferred from mouse homologs in Zhu et al. 2012). Thus, TBX21 restricts naive CD4⁺ T cells to the Th1 fate.

TBX21 then directly activates expression of Th1 regulators and cell markers, including C-X-C chemokine receptor type 3 (CXCR3) (inferred from mouse homologs in Szabo et al. 2000, Lord et al. 2005, Zhu et al. 2012, Gökmen et al. 2013), Tumor necrosis factor (TNF, TNFA) (Jenner et al. 2009, and inferred from mouse homologs in Zhu et al. 2012), C-C motif chemokine 3 (CCL3) (Jenner et al. 2009, and inferred from mouse homologs in supplement-02 of Zhu et al. 2012). TBX21 also upregulates expression of Protein-tyrosine sulfotransferase 2 (TPST2) (inferred from mouse homologs in Lord et al. 2005), which is linked to the ability of Th1 cells to migrate and access inflamed skin (Pritchard et al. 2019).

References

Pritchard GH, Kedl RM & Hunter CA (2019). The evolving role of T-bet in resistance to infection. Nat Rev Immunol, 19, 398-410. [🔗](#)

Schulz EG, Mariani L, Radbruch A & Höfer T (2009). Sequential polarization and imprinting of type 1 T helper lymphocytes by interferon-gamma and interleukin-12. Immunity, 30, 673-83. [🔗](#)

Zhong L, Wang YH, Kahlfuss S, Jishage M, McDermott M, Yang J, ... Feske S (2025). STIM1-mediated NFAT signaling synergizes with STAT1 to control T-bet expression and T_H1 differentiation. Nat Immunol, 26, 484-496. [🔗](#)

Wei L, Vahedi G, Sun HW, Watford WT, Takatori H, Ramos HL, ... Kanno Y (2010). Discrete roles of STAT4 and STAT6 transcription factors in tuning epigenetic modifications and transcription during T helper cell differentiation. Immunity, 32, 840-51. [🔗](#)

Fang D, Cui K, Cao Y, Zheng M, Kawabe T, Hu G, ... Zhu J (2022). Differential regulation of transcription factor T-bet induction during NK cell development and T helper-1 cell differentiation. Immunity, 55, 639-655.e7. [🔗](#)

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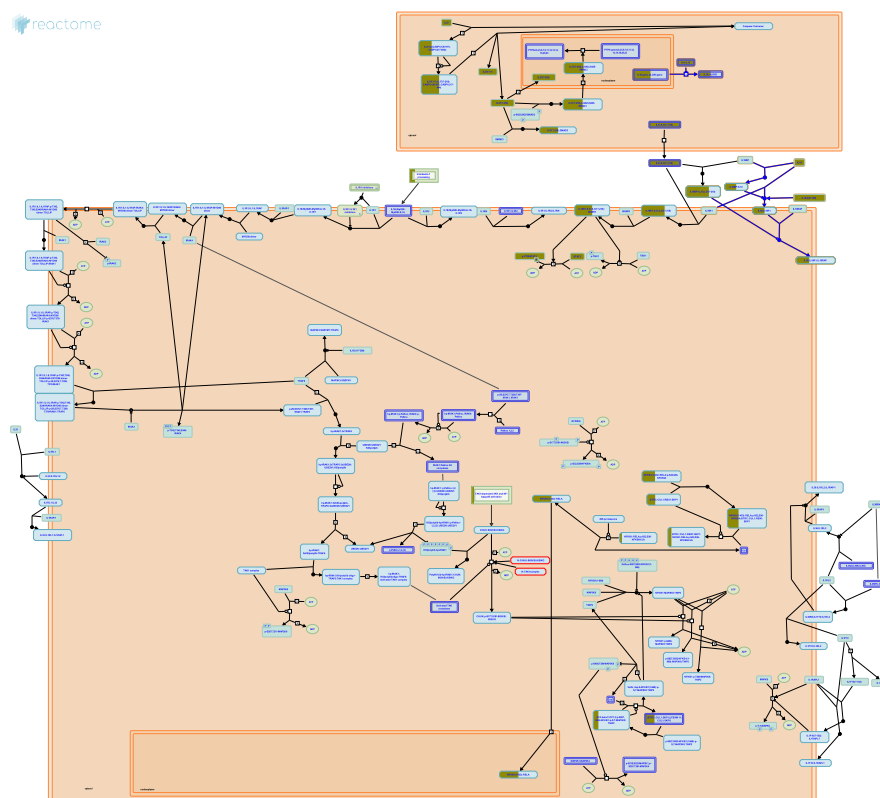
Date	Action	Author
2025-03-17	Edited	May B
2025-03-17	Authored	May B
2025-03-17	Created	May B
2025-08-04	Reviewed	Barbutti I
2025-08-05	Reviewed	Lord GM
2026-01-08	Modified	May B

5 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id	Input	UniProt Id
IFNG	P01579	IL12RB2	Q99665
STAT1	P42224-1	STAT4	Q14765

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IFNG	ENSG00000111537	IL12RB2	ENSG00000081985	IL4	ENSG00000113520

24. Interleukin-18 signaling (R-HSA-9012546)



Interleukin-18 (IL18, pro-IL18) is a pleiotropic and pro inflammatory cytokine. It belongs to the Interleukin-1 (IL1 superfamily (Alboni et al. 2010, Krumm et al. 2017, Dinarello 1999). IL18 is synthesized as an inactive 24-kDa precursor protein that is cleaved by extracellular proteases such as caspase-1, protease 3, serine protease, elastase or cathepsin G (Fantuzzi & Dinarello 1999, Gracie et al. 2004, Sugawara et al. 2001), forming an 18-kDa mature protein (Arend et al. 2008, Akita et al. 1997, Fantuzzi et al. 1998, Ghayur et al. 1997, Gu et al. 1997, Ushio et al. 1996).

IL18 also occurs as a short isoform, the result of an alternative splicing event that removes 57 bp/19 aa (IL18alpha) (Conti et al. 1997, Yang et al. 2005). This short isoform has a modest synergistic action with the IL18 canonical active form.

The IL18 receptor (IL18R) belongs to the Interleukin-1 receptor/Toll like receptor superfamily. It consists of two subunits, Interleukin-18 receptor 1 (IL18R1, IL-18R α , IL1Rrp1, IL18R1, IL-1R5) and Interleukin-18 receptor accessory protein (IL18RAP, IL18RB, IL-18R β , IL-18RacP, IL-18RII or IL-1R7). Both subunits have three extracellular immunoglobulin-like domains and one intracellular Toll/IL-1 receptor (TIR) domain (O'Neill & Dinarello 2000, Sims 2002). It is believed that IL18 binds first to IL18R1 and later recruits IL18RAP to form a high-affinity heterotrimeric complex (Sims 2002, Sergi & Pentilla 2004, Alboni et al. 2009). A short isoform of IL18R1 lacks the TIR domain (IL18R1 type II) (Alboni et al. 2009), which is required for signaling, leading to the suggestion that IL18R1 type II is a decoy receptor (Colotta et al. 1994). A truncated form of IL18RAP containing only one of the three immunoglobulin domains stabilizes IL18 binding to IL18R1 but prevents signaling.

IL-18 binding protein (IL18BP), a 38-kDa soluble protein, is another negative regulator of IL18 signaling. It has some sequence homology with IL18R1 (Im et al. 2002, Kim et al. 2002, Novick et al. 1999). IL18BP binds with high affinity to mature IL18, preventing its interaction with IL18R1. Several isoforms IL18BP have been described (Kim et al. 2000). Interleukin-37 (IL37, IL-1F7), another negative regulator of IL18 signaling, is able to bind IL18BP and IL18RAP preventing signaling (Buller et al. 2002, Pan et al. 2001, Kumar et al. 2002).

IL18 stimulates Interferon gamma (IFNG, IFN- γ) production from T-helper lymphocytes cells (Th1) and macrophages and enhances the cytotoxicity of natural killer (NK) cells. IL18 stimulated IFNG production is synergistically amplified by other Th1-related cytokines such as IL2, IL15, IL12 and IL23 (Boraschi & Dinarello 2006, Park et al. 2007, Dinarello 2007, Dinarello & Fantuzzi 2003).

References

Alboni S, Cervia D, Sugama S & Conti B (2010). Interleukin 18 in the CNS. J Neuroinflammation, 7, 9 . 

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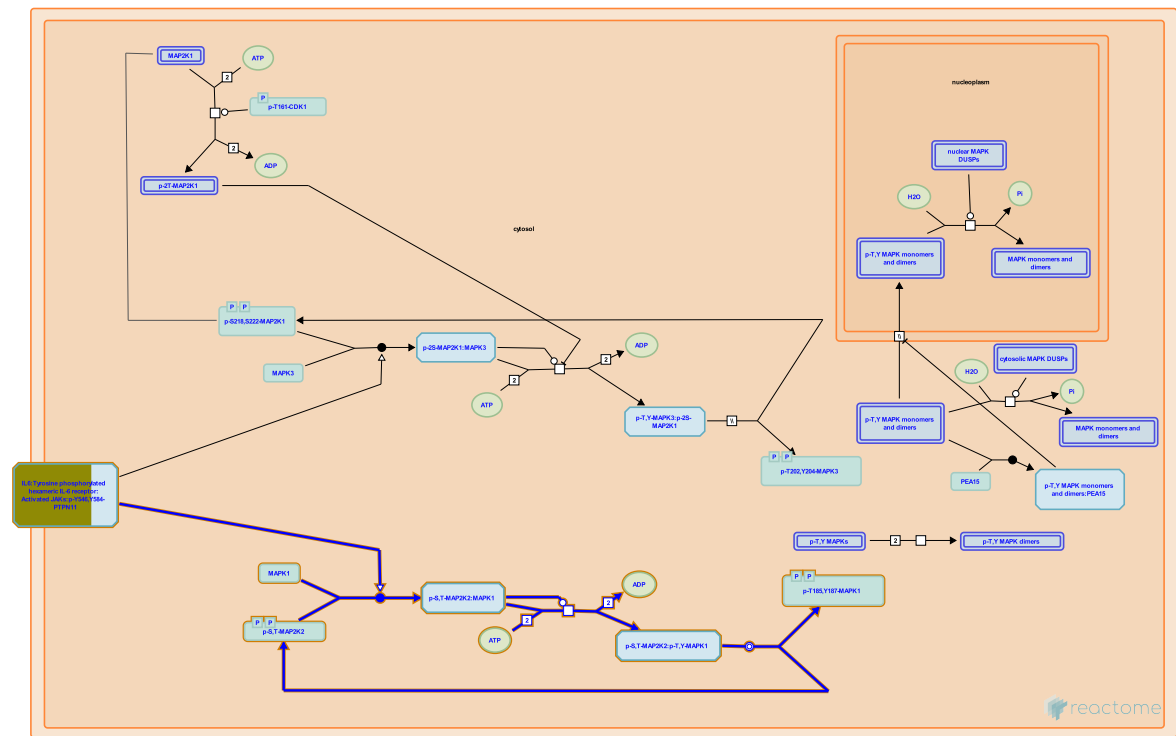
Date	Action	Author
2016-01-28	Reviewed	Meldal BH
2017-07-17	Created	Duenas C
2017-08-08	Edited	Duenas C
2017-08-08	Authored	Duenas C
2026-03-16	Modified	Weiser JD

4 submitted entities found in this pathway, mapping to 6 Reactome entities

Input	UniProt Id	Input	UniProt Id
IL13	P35225	IL18	Q14116, Q14116-2
IL37	Q9NZH6	IL4	P05112

Input	Ensembl Id
IL18	ENSG00000150782

25. MAPK1 (ERK2) activation (R-HSA-112411)



Cellular compartments: cytosol, nucleoplasm.

Mitogen-activated protein kinase kinase MAP2K2 (also known as MEK2) is a dual threonine and tyrosine recognition kinase that phosphorylates and activates MAPK1 (ERK2) (Ohren et al. 2004; Roskoski 2012).

References

Zheng CF & Guan KL (1994). Activation of MEK family kinases requires phosphorylation of two conserved Ser/Thr residues. EMBO J, 13, 1123-31. [🔗](#)

Roskoski R Jr (2012). MEK1/2 dual-specificity protein kinases: structure and regulation. Biochem. Biophys. Res. Commun., 417, 5-10. [🔗](#)

Edit history

Date	Action	Author
2004-04-29	Created	Charalambous M
2005-02-04	Authored	Charalambous M
2007-11-08	Reviewed	Greene LA
2026-03-12	Edited	Schmidt EE

6 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IL11RA	P08887	IL6	P05231	IL6R	P08887, P08887-2
IL6ST	P40189-1	JAK1	P23458	TYK2	P29597

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

100 of the submitted entities were found, mapping to 146 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABCB1	P21439	ACTB	P60709, P63261	ACTC1	P68032
ATG12	O94817	ATG16L1	Q676U5	ATG5	Q9H1Y0
ATP6V0C	P27449	BTRC	Q9Y297	CARD8	Q9C000
CEP152	O94986	CEP192	Q8TEP8	CFL1	P23528
CHAT	P28329	CNTFR	P26992	CREBBP	Q92793
CSF2	P04141	CTLA4	P16410	CYTH1	Q15438, Q99418
CYTH2	Q15438, Q99418	CYTH3	O43739	E2F6	O75461
ERBIN	Q96RT1	FGFR3	P22607-1, P22607-2	GFOD1	Q9NXC2
GRIN2C	Q14957	GSN	P06396	IFNA1	P01563, P01569, P01570, P0DY56
IFNA2	P01563	IFNAR1	P17181	IFNAR2	P48551-2
IFNB1	P01574	IFNG	P01579	IFNGR1	P15260, P17181
IFNL1	Q8IU54	IFNL2	Q8IZI9, Q8IZJ0	IFNL3	Q8IZI9, Q8IZJ0
IFNLR1	Q8IU57, Q969J5	IL10	Q9NPH9	IL10RA	Q8IU57
IL10RB	Q08334	IL11RA	P08887	IL12A	P29459
IL12B	P29460	IL12RB1	P42701	IL12RB2	Q99665
IL13	P35225	IL13RA1	P78552	IL13RA2	Q14627
IL17A	Q16552	IL18	Q14116	IL19	Q9UHD0
IL20	Q9NYY1	IL20RA	Q9UHF4	IL20RB	Q6UXL0
IL21R	Q9HBE5	IL22	Q9GZX6	IL22RA1	Q8N6P7
IL22RA2	Q8N6P7, Q969J5	IL23A	Q9NPF7	IL23R	Q5VWK5
IL24	Q13007	IL26	Q9NPH9	IL27RA	Q6UWB1
IL2RA	P01589	IL2RB	P14784	IL37	Q9NZH6
IL4	P05112	IL4R	P24394	IL5RA	Q01344
IL6	P05231	IL6R	P08887, P08887-2	IL6ST	P40189, P40189-1, P40189-2
IL9	P15248	IL9R	Q01113	IRF5	Q13568
JAK1	P23458	LGALS3	P17931	NLRP3	Q96P20
NOD2	Q9HC29	OPTN	Q96CV9	PRLR	Q5VWK5
RAB33A	Q14088	RAB33B	Q9H082	RALGAPB	Q86X10
RB1CC1	Q8TDY2	RELA	Q04206	RIPK2	O43353
RNF2	Q99496	SGF29	Q96ES7	SGTA	O43765
STAT1	P42224	STAT3	P40763	STAT4	Q14765
TRIM27	P14373	TRIM28	Q13263	TYK2	P29597
UBQLN1	Q9UMX0	VCP	P55072	VIM	P08670

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ACTB	ENST00000331789	ATP6V0C	ENSG00000185883	CFL1	ENSG00000172757
CSF2	ENSG00000164400	CTLA4	ENSG00000163599	IFNA13	ENSG00000233816
IFNA2	ENSG00000188379	IFNB1	ENSG00000171855	IFNG	ENSG00000111537
IL10	ENSG00000136634	IL10RA	ENSG00000110324	IL12A	ENSG00000168811
IL12B	ENSG00000113302	IL12RB2	ENSG00000081985	IL13	ENSG00000169194
IL17A	ENSG00000112115	IL18	ENSG00000150782	IL23A	ENSG00000110944

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IL23R	ENSG00000162594	IL2RA	ENSG00000134460	IL4	ENSG00000113520
IL4R	ENSG00000077238	IL6	ENSG00000136244	IL6R	ENSG00000160712
IRF5	ENSG00000128604	LGALS3	ENSG00000131981	NLRP3	ENSG00000162711
STAT1	ENSG00000115415	STAT3	ENSG00000168610	VIM	ENSG00000026025

7. Identifiers not found

These 10 identifiers were not found neither mapped to any entity in Reactome.

BAHD1	BCL6B	CARD10	CEP170	FAM167A	IFNE	IFNK	IFNW1
INAVA	IRGM						